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OVERVIEW OF THE INTEGUMENTARY SYSTEM

The **skin (cutis, integument)** and its derivatives constitute the **integumentary system**. The skin forms the external covering of the body and is its largest organ, constituting 15% to 20% of its total mass. The skin consists of two main layers:

- The **epidermis** is composed of a **keratinized stratified squamous epithelium** that grows continuously but maintains its normal thickness by the process of **desquamation**. Epidermis is derived from **ectoderm**.
- The **dermis** is composed of a **dense connective tissue** that imparts mechanical support, strength, and thickness to the skin. Dermis is derived from **mesoderm**.

The **hypodermis** contains variable amounts of adipose tissue arranged into lobules separated by connective tissue septa. It lies deep to the dermis and is equivalent to the **subcutaneous fascia** described in gross anatomy. In well-nourished individuals and in individuals living in cold climates, the adipose tissue can be quite thick.

The **epidermal derivatives of the skin** (epithelial skin appendages) include the following structures and integumentary products:

- **Hair follicles** and **hair**
- **Sweat (sudoriferous) glands**

- **Sebaceous glands**
- **Nails**
- **Mammary glands**

The integumentary system performs essential functions related to its external surface location.

Skin and its derivatives constitute a complex organ composed of many different cell types. The diversity of these cells and their ability to work together provide many functions that allow the individual to cope with the external environment. Major functions of the skin include the following:

- It acts as a **barrier** that protects against physical, chemical, and biologic agents in the external environment (i.e., mechanical barrier, permeability barrier, ultraviolet barrier).
- It provides **immunologic information** obtained during antigen processing to the appropriate effector cells in the lymphatic tissue.
- It participates in **homeostasis** by regulating body temperature and water loss.
- It conveys **sensory information** about the external environment to the nervous system.
- It performs **endocrine functions** by secreting hormones, cytokines, and growth factors and converting precursor molecules into hormonally active molecules (vitamin D₃).
- It functions in **excretion** through the exocrine secretion of sweat, sebaceous, and apocrine glands.

In addition, certain lipid-soluble substances may be absorbed through the skin. Although not a function of skin, this property is frequently used to deliver therapeutic agents. For example, nicotine, steroid hormones, and seasickness medications are frequently delivered through the skin in the form of small sticking plasters or patches. To reduce **nicotine withdrawal symptoms** during smoking cessation, nicotine patches are often used to provide a small constant dose of nicotine without the dangerous effects of tobacco smoke.

Skin is categorized as thick or thin, a reflection of thickness and location.

The thickness of the skin varies over the surface of the body, from less than 1 mm to more than 5 mm. However, the skin is obviously both grossly and histologically different at two locations: **the palms of the hands and the soles of the feet**. These areas are subject to the most abrasion, are hairless, and have a much thicker epidermal layer than skin in any other location. This hairless skin is referred to as **thick skin**. Elsewhere, the skin possesses a much thinner epidermis and is called **thin skin**. It contains hair follicles in all but a few locations.

The terms thick skin and thin skin, as used in histologic description, are misnomers and refer only to the thickness of the epidermal layer. Anatomically, the thickest skin is found on the upper portion of the back where the dermis is exceedingly thick. The epidermis of the upper back, however, is comparable to that of thin skin found elsewhere on the body. In contrast, in certain other sites such as the eyelid, the skin is extremely thin.

LAYERS OF THE SKIN

Epidermis

The **epidermis** is composed of **stratified squamous epithelium** in which four distinct layers can be identified. In the case of thick skin, a fifth layer is observed (Figs. 15.1 and 15.2). Beginning with the deepest layer, these are as follows:

- the **stratum basale**, also called the **stratum germinativum** because of the presence of mitotically active cells, the stem cells of the epidermis;
- the **stratum spinosum**, also called the **spinous layer** or **prickle cell layer** because of the characteristic light microscopic appearance of short processes extending from cell to cell;
- the **stratum granulosum**, which contains numerous intensely staining granules;
- the **stratum lucidum**, which is limited to thick skin and considered a subdivision of the stratum corneum; and
- the **stratum corneum**, which is composed of keratinized cells.

Differentiation of epithelial cells constitutes a specialized form of apoptosis.

Terminal differentiation of the epidermal cells, which begins with the cell divisions in the stratum basale, is considered a specialized form of apoptosis. Cells in the stratum granulosum exhibit typical apoptotic nuclear morphology, including fragmentation of their DNA. However, the cellular fragmentation associated with normal apoptosis does not occur; instead, the cells become filled with filaments of the intracellular protein **keratin** and are later sloughed from the skin surface.

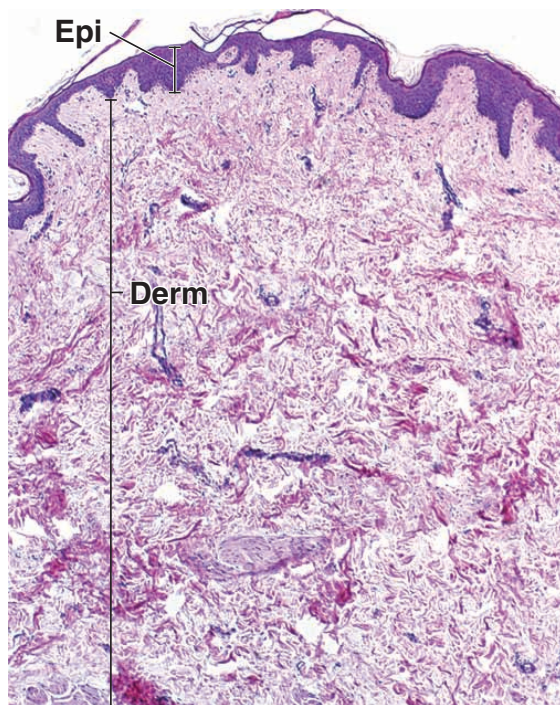


FIGURE 15.1 ▲ Photomicrograph showing the layers of thin skin. This hematoxylin and eosin (H&E)-stained specimen from human skin shows the two chief layers of the skin—the epidermis (*Epi*) and dermis (*Derm*). The epidermis forms the surface; it consists of stratified squamous epithelium that is keratinized. The dermis consists of two layers: the papillary layer, which is the most superficial layer and is adjacent to the epidermis, and the more deeply positioned reticular layer. The boundary between these two layers is not conspicuous; the papillary layer is, however, more cellular than the reticular layer. In addition, the collagen fibers of the reticular layer are thick (clearly visible in the lower part of the figure); those of the papillary layer are thin. $\times 45$.

The stratum basale provides for epidermal cell renewal.

The **stratum basale** is represented by a single layer of cells that rests on the basal lamina (Plate 42, page 514). It contains the **stem cells** from which new cells, the **keratinocytes**, arise by mitotic division. For this reason, the stratum basale is also called the **stratum germinativum**. The cells are small and cuboidal to low columnar. They have less cytoplasm than the cells in the layer above; consequently, their nuclei are more closely spaced. The closely spaced nuclei, in combination with the basophilic cytoplasm of these cells, impart a noticeable basophilia to the stratum basale. The basal cells also contain various amounts of melanin (described later) in their cytoplasm that is transferred from neighboring melanocytes interspersed in this layer. Basal cells exhibit extensive cell junctions; they are connected to each other and to keratinocytes by desmosomes and to the underlying basal lamina by hemidesmosomes. As new keratinocytes arise in this layer by mitotic division, they move into the next layer, thus beginning their process of upward migration. This process terminates when the cell becomes a mature keratinized cell, which is eventually sloughed off at the skin surface.

The cells of the stratum spinosum characteristically exhibit spinous processes.

The **stratum spinosum** is at least several cells thick. **Keratinocytes** in this layer are larger than those of the stratum

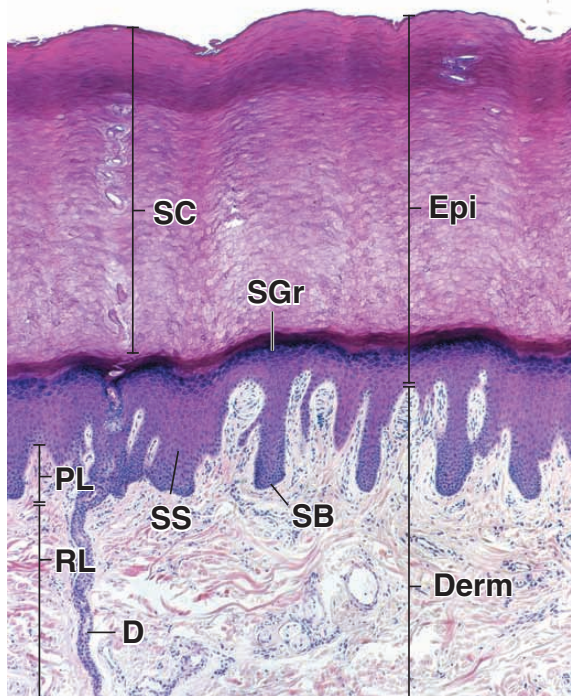


FIGURE 15.2 ▲ Photomicrograph showing the layers of thick skin. This specimen obtained from the skin of the sole of the foot (human) shows epidermis (Epi) containing the extremely thick stratum corneum (SC). Remaining layers of the epidermis (except for the stratum lucidum, which is not present on this slide)—that is, the stratum basale (SB), the stratum spinosum (SS), and the stratum granulosum (SGr)—are clearly visible in this routine H&E preparation. The duct of a sweat gland (D) can be seen on the left as it traverses the dermis (Derm) and further spirals through the epidermis. At the sites where the ducts of the sweat gland enter the epidermis, note the epidermal downgrowths known as *interpapillary pegs*. The dermis contains papillae, protrusions of connective tissue that lie between the interpapillary pegs. Note also the greater cellularity of the papillary layer (PL) and that the collagen fibers of the reticular layer (RL) are thicker than those of the papillary layer. $\times 65$.

basale. They exhibit numerous **cytoplasmic processes** or spines, which gives this layer its name (Fig. 15.3 and Plate 42, page 514). The processes are attached to similar processes of adjacent cells by **desmosomes**. In the light microscope, the site of the desmosome appears as a slight thickening called the *node of Bizzozero*. The processes are usually conspicuous, in part because the cells shrink during preparation and a resultant expanded intercellular space develops between the spines. Because of their appearance, the cells that constitute this layer are often referred to as **prickle cells**. As the cells mature and move to the surface, they increase in size and become flattened in a plane parallel to the surface. This arrangement is particularly notable in the most superficial spinous cells, where the nuclei also become elongate instead of ovoid, matching the acquired squamous shape of the cells.

The cells of the stratum granulosum contain conspicuous keratohyalin granules.

The **stratum granulosum** is the most superficial layer of the nonkeratinized portion of the epidermis. This layer varies from one to three cells thick. **Keratinocytes** in this layer contain numerous **keratohyalin granules**, hence

the name of the layer. These granules contain cystine-rich and histidine-rich proteins, which are the precursors of the protein **filaggrin**, which aggregates the **keratin filaments** present within the cornified cells of the stratum corneum. Keratohyalin granules are irregular in shape and variable in size. Because of their intense basophilic staining, they are readily seen in routine histologic sections.

The stratum corneum consists of anucleate squamous cells largely filled with keratin filaments.

Usually, an abrupt transition occurs between the nucleated cells of the stratum granulosum and the flattened, desiccated, anucleate cells of the **stratum corneum**. The cells in the stratum corneum are the most differentiated cells in the skin. They lose their nucleus and cytoplasmic organelles and become filled almost entirely with keratin filaments. The thick plasma membrane of these cornified, keratinized cells is coated from the outside, in the deeper portion of this layer, with an extracellular layer of lipids that form the major constituent of the **water barrier** in the epidermis.

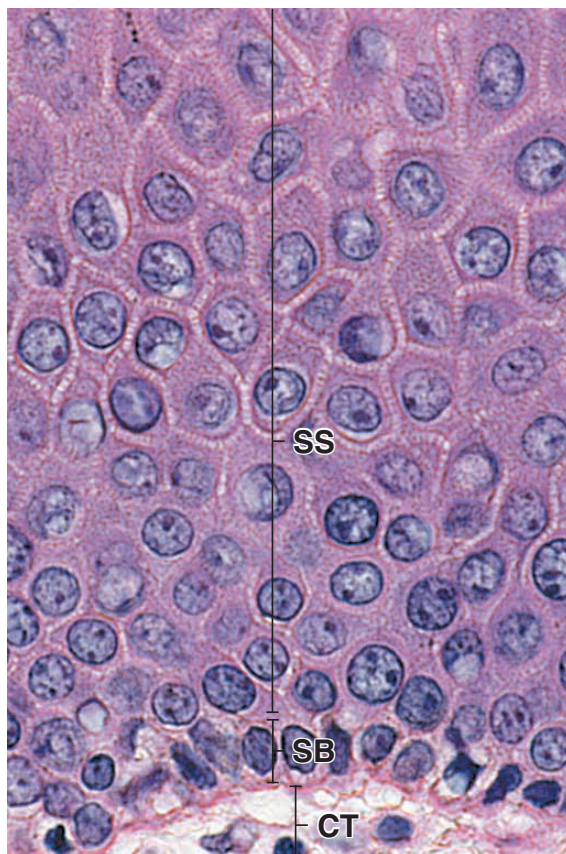


FIGURE 15.3 ▲ Photomicrograph of the stratum spinosum and stratum basale. The epidermis of thin skin is shown here at higher magnification. The one-cell-deep layer at the base of the epidermis just above the connective tissue (CT) of the dermis is the stratum basale (SB). The cells of this layer rest on the basement membrane. A layer referred to as the *stratum spinosum* (SS) is located just above the stratum basale. It consists of cells with spinous processes on their surfaces. These processes are attached to spinous processes of neighboring cells by desmosomes and together appear as intercellular bridges. $\times 640$.

The stratum corneum is the layer that varies most in thickness, being thickest in thick skin. The thickness of this layer constitutes the principal difference between the epidermis of thick and thin skin. This cornified layer will become even thicker at sites subjected to unusual amounts of friction, as in the formation of calluses on the palms of the hands and on the fingertips.

The **stratum lucidum**, considered a subdivision of the stratum corneum by some histologists, is normally only well seen in thick skin. In the light microscope, it often has a refractile appearance and may stain poorly. This highly refractile layer contains eosinophilic cells in which the process of keratinization is well advanced. The nucleus and cytoplasmic organelles become disrupted and disappear as the cell gradually fills with keratin.

Dermis

Attachment of epidermis to dermis is enhanced by an increased interface between the two tissues.

The junction between the epidermis and dermis (**epidermal–dermal junction**) is seen in the light microscope as an uneven boundary except in the thinnest skin. Sections of skin cut perpendicular to the surface reveal numerous finger-like connective tissue protrusions, **dermal papillae**, that project into the undersurface of the epidermis (see Figs. 15.1 and 15.2). The papillae are complemented by what appear to be similar epidermal protrusions, called **epidermal ridges** or **rete ridges**, that project into the dermis. If the plane of section is parallel to the surface of the epidermis and passes at a level that includes the dermal papillae, however, the epidermal tissue appears as a continuous sheet of epithelium, containing circular islands of connective tissue within it. The islands are cross-sections of true finger-like dermal papillae that project into the epidermis. At sites where increased **mechanical stress** is placed on the skin, the epidermal ridges are much deeper (the epithelium is

thicker), and the dermal papillae are much longer and more closely spaced, creating a more extensive interface between the dermis and epidermis. This phenomenon is particularly well demonstrated in histologic sections that show both palmar and dorsal surfaces of the hand, as in a section of a finger.

True dermal ridges are present in thick skin in addition to dermal papillae.

Dermal ridges tend to have a parallel arrangement, with the dermal papillae located between them. These ridges form a distinctive pattern that is genetically unique to each individual and is reflected in the appearance of epidermal grooves and ridges on the surface of the skin. These patterns are the basis of the science of **dermatoglyphics** or fingerprint and footprint identification.

The dermal ridges and papillae are most prominent in the thick skin of the palmar and plantar surfaces. Here, the basal surface of the epidermis greatly exceeds its free surface. The germinal layer is thus spread over a large area; assuming a near-constant rate of mitosis in the stratum germinativum, more cells per unit time enter the stratum corneum in thick skin than in thin skin. These additional cells are thought to account for the greater thickness of the cornified layer in thick skin.

Hemidesmosomes strengthen the attachment of the epidermis to the underlying connective tissue.

When studied with the transmission electron microscope (TEM), the basal surface of the basal epidermal cells exhibits a pattern of irregular cytoplasmic protrusions that increase the attachment surface between the epithelial cell and its subjacent basal lamina. A series of **hemidesmosomes** links the intermediate filaments of the cytoskeleton into the basal lamina. In addition, **focal adhesions** that anchor actin filaments into the basal lamina are also present. These specialized anchoring junctions are discussed on pages 141–143.

FOLDER 15.1 Clinical Correlation: Cancers of Epidermal Origin

Three major types of **skin cancer** originate from cells in the epidermis. In general, skin cancer is caused by unprotected, long-term exposure to the sun's ultraviolet radiation. The most common type is the **basal cell carcinoma**, which microscopically, as its name implies, resembles cells from the stratum basale of the epidermis. Basal cell carcinoma is a slow-growing tumor that usually does not metastasize. Typically, the cancer cells arise from the follicular bulge of the external root sheath of the hair follicle. Almost in all cases of basal cell carcinoma, the recommended treatment is surgical removal of the tumor.

The second most common skin cancer is the **squamous cell carcinoma** with more than 200,000 cases each year. Individuals with this form of cancer usually develop a small painless nodule or patch that is surrounded by an area of inflammation. Squamous cell carcinoma is characterized by highly atypical cells at all levels of the epidermis (carcinoma in situ). Disruption of the basement membrane results in spread (metastasis) of tumor cells to the lymph nodes. Squamous cell carcinoma is known for variable differentiation patterns ranging from polygonal squamous cells arranged in orderly

lobules and zones of keratinization to rounded cells with foci of necrosis and occasional single keratinized cells. Treatment for squamous cell carcinoma depends on histological type, size, and location. It may include surgical excision, curettage and electrodesiccation, cryotherapy (freezing with liquid nitrogen), or chemo- or radiotherapy. Moh's micrographic surgical procedure is being used for locally recurrent skin cancers. This procedure involves shaving away one-by-one thin layers of epidermis and examining them under a microscope for the presence of malignant cells. When the shaving is cancer-free, the surgery is complete. This method preserves as many unaffected skin layers as possible while making certain that all cancer cells are removed.

Malignant melanoma is the most serious form of skin cancer if not recognized at an early stage and surgically removed. Individual melanoma cells, which originate from melanocytes, contain large nuclei with irregular contours and prominent eosinophilic nucleoli. These cells either aggregate in nests or are scattered through the entire thickness of the epidermis (Fig. F15.1.1). They may reside only in the epidermis (melanoma in situ) or extend into the underlying papillary

(continues on page 492)

FOLDER 15.1 Clinical Correlation: Cancers of Epidermal Origin (continued)

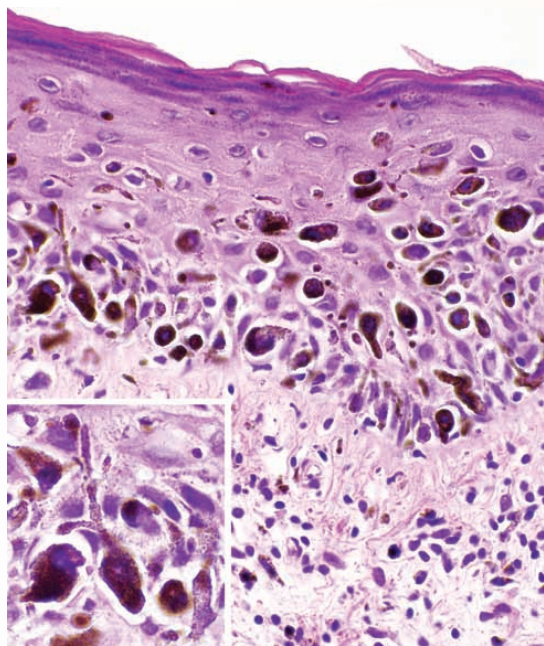


FIGURE F15.1.1 ▲ Photomicrograph of a malignant melanoma lesion in the early stage of radial growth phase. This section of the skin shows a layer of the epidermis containing atypical (hyperplastic) cells loaded with dark-brown pigment granules containing melanin. These cells represent atypical melanocytes that normally should reside in the stratum basale of the epidermis. At this stage of disease, these abnormal melanocytes migrate to the upper layers of the epidermis (melanocytic hyperplasia). There are scattered small nests of atypical cells in the dermis. Note the accumulation of lymphocytes in the superficial dermis. $\times 320$. Inset shows enlarged nest containing melanocytes with clearly visible processes containing melanin granules. $\times 640$.

layer of the dermis. With time, the melanoma undergoes a **radial growth phase**. The melanocytes grow in all directions, upward in the epidermis, downward into the dermis, and peripherally in the epidermis. At this early stage, the melanoma tends not to metastasize. On the skin surface, it presents itself as an irregularly pigmented multicolor lesion, appearing black with parts brown to light brown, and a mixture of pink to red or shades of blue (Fig. F15.1.2). In time (approximately 1 or 2 years), melanocytes exhibit mitotic

activity and form round nodules growing perpendicularly to the surface of the skin. In this **vertical growth phase**, the melanocytes display little or no pigment and usually metastasize into regional lymph nodes.

The **ABCD rule** is helpful for remembering the signs and symptoms of melanoma (see Fig. F15.1.2):

- **A**symmetrical shape of skin lesion
- **B**order of the lesion is irregular
- **C**olor variations; melanomas usually have multiple colors
- **D**iameter of skin lesion; moles greater than 6 mm are more likely to be suspicious

Surgery is a choice of treatment for localized malignant melanoma of the skin. A multidisciplinary approach is used for advanced malignant melanoma, including surgery combined with chemotherapy or immunotherapy with adjuvant treatment.



FIGURE F15.1.2 ▲ Photograph of the skin with malignant melanoma during radial growth phase. In this individual, malignant melanoma is represented by the relatively flat, irregularly pigmented multicolor lesion. The largest nodule appears *ebony black*. It is adjacent to a slightly elevated region that is colored in shades from *dark brown* to *light brown* with two *reddish* smaller nodules in *pink* coloration. At this early stage, melanocytes grow in all directions, upward in the epidermis, downward into the dermis, and peripherally in the epidermis. (Reproduced from Storm CA, Elder DE. The Skin. In: Rubin R, Strayer DS (eds): Rubin's Pathology: Clinicopathologic Foundations of Medicine, 5th ed. Baltimore: Lippincott Williams & Wilkins, 2008.)

The dermis is composed of two layers: the papillary layer and the reticular layer.

Examination of the full thickness of the dermis at the light microscope level reveals two structurally distinct layers.

- The **papillary layer**, the more superficial layer, consists of loose connective tissue immediately beneath the epidermis (Plate 43, page 516). The collagen fibers located in this part of the dermis are not as thick as those in the deeper portion. This delicate collagen network contains predominately type I and type III collagen molecules. Similarly, the elastic fibers

- are thread-like and form an irregular network. The papillary layer is relatively thin and includes the substance of the dermal papillae and dermal ridges. It contains blood vessels that serve but do not enter the epidermis. It also contains nerve processes that either terminate in the dermis or penetrate the basal lamina to enter the epithelial compartment. Because the blood vessels and sensory nerve endings are concentrated in this layer, they are particularly apparent in the dermal papillae.
- The **reticular layer** lies deep to the papillary layer. Although its thickness varies in different parts of the body, it is always considerably thicker and less cellular than the papillary layer.

It is characterized by thick, irregular bundles of mostly type I collagen and by coarser elastic fibers. The collagen and elastic fibers are not randomly oriented but form regular lines of tension in the skin called **Langer's lines**. Skin incisions made parallel to Langer's lines heal with the least scarring.

In the skin of the areolae, penis, scrotum, and perineum, smooth muscle cells form a loose plexus in the deepest parts of the reticular layer. This arrangement accounts for the puckering of the skin at these sites, particularly in erectile organs.

Layers of adipose tissue, smooth muscle, and, in some sites, striated muscle may be found just beneath the reticular layer.

Deep to the reticular layer is a **layer of adipose tissue**, the **panniculus adiposus**, which varies in thickness. This layer serves as a major energy storage site and also provides insulation. It is particularly thick in individuals who live in cold climates. This layer and its associated loose connective tissue constitute the **hypodermis** or **subcutaneous fascia**.

Individual smooth muscle cells or small bundles of smooth muscle cells that originate in this layer form the **arrector pili muscles** that connect the deep part of hair follicles to the more superficial dermis. Contraction of these muscles in humans produces the erection of hairs and puckering of skin called "goose flesh." In other animals, the erection of hairs serves in both thermal regulation and fright reactions.

A thin layer of striated muscle, the **panniculus carnosus**, lies deep to the subcutaneous fascia in many animals. Although largely vestigial in humans, it remains well defined in the skin of the neck, face, and scalp, where it constitutes the **platysma** muscle and the other **muscles of facial expression**.



CELLS OF THE EPIDERMIS

There are four different types of cells in the epidermis:

- **Keratinocytes** are highly specialized epithelial cells designed to perform a very specific function: separation of the organism from its external environment. They account for 85% of the cells in the epidermis.
- **Melanocytes** are the pigment-producing cells of the epidermis. They account for approximately 5% of the cells in the epidermis.
- **Langerhans' cells** are the antigen-presenting cells involved in signaling in the immune system. They account for approximately 2% to 5% of the cells in the epidermis.
- **Merkel's cells** are the sensitive mechanoreceptor cells associated with sensory nerve endings. They account for approximately 6% to 10% of the cells in the epidermis.

Keratinocytes

The **keratinocyte** is the predominant cell type of the epidermis. These cells originate in the stratum basale of the epidermis. On leaving this layer, keratinocytes assume two essential activities:

- They produce **keratins (cytokeratins)**, major heteropolymeric structural proteins of the epidermis (see Table 2.3, page 62). Keratins form intermediate filaments; they constitute almost 85% of fully differentiated keratinocytes.
- They participate in the formation of the **epidermal water barrier**.

The keratinocytes in the basal layer contain numerous free ribosomes, scattered 7- to 9-nm intermediate (keratin) filaments, a small Golgi apparatus, mitochondria, and rough-surfaced endoplasmic reticulum (rER). The cytoplasm of immature keratinocytes appears basophilic in histologic sections because of the large number of free ribosomes, most of which are engaged in the synthesis of keratins, which will later be assembled into **keratin filaments**. These filaments are classified as intermediate filaments, although they are more commonly called **tonofilaments**.

As the cells enter and are moved through the stratum spinosum, the synthesis of keratin filaments continues, and the filaments become grouped into bundles sufficiently thick to be visualized in the light microscope. These bundles are called **tonofibrils**. The cytoplasm becomes eosinophilic because of the staining reaction of the tonofibrils that fill more and more of the cytoplasm.

Keratohyalin granules contain intermediate filament-associated proteins that aid in the aggregation of keratin filaments.

In the upper part of the stratum spinosum (Fig. 15.4), the free ribosomes within the keratinocytes begin to synthesize **keratohyalin granules** that become the distinctive feature of the cells in the stratum granulosum (Plate 42, page 514). Keratohyalin granules contain the two major intermediate filament-associated proteins, **filaggrin** and **trichohyalin**. The appearance of the granules and expression of filaggrin in the keratinocytes are often used as a clinical marker for the initiation of the final stage of apoptosis. As the number of granules increases, the contents of the granules are released into the keratinocyte cytoplasm. Filaggrin and trichohyalin function as promoters in the aggregation of keratin filaments into tonofibrils, thus initiating the conversion of granular cells into cornified cells. This process is called **keratinization** and occurs in 2 to 6 hours, the time it takes for the cells to leave the stratum granulosum and enter the stratum corneum. The keratin fibril formed in this process is called **soft keratin** in contrast to the **hard keratin** of hair and nails (see below).

The transformation of a granular cell into a keratinized cell also involves breakdown of the nucleus and other organelles and thickening of the plasma membrane. This is accompanied by a change in pH, which decreases from approximately neutral (pH 7.17) in the stratum granulosum to acidic at the surface of the stratum corneum, ranging between pH 4.5 and 6.0.

Desquamation of surface keratinocytes from the stratum corneum is regulated by proteolytic degradation of the cells' desmosomes.

Cells are regularly exfoliated or desquamated from the surface of the stratum corneum. The **continuous exfoliation** of surface keratinocytes is a regulated proteolytic process that involves degradation of the cells' desmosomes. The human **kallikrein-related serine peptidases** such as **KLK5**, **KLK7**, and **KLK14** cause cleavage of desmosomes in a pH-dependent manner. A physiological serine protease inhibitor, **lymphoepithelial Kazal-type inhibitor (LEKTI)**, through its interactions with KLKs in neutral pH, prevents desmosomal cleavage. However, as pH decreases in more superficial portions of the stratum corneum as described, LEKTI progressively releases KLKs at the lower pH, thus allowing KLKs

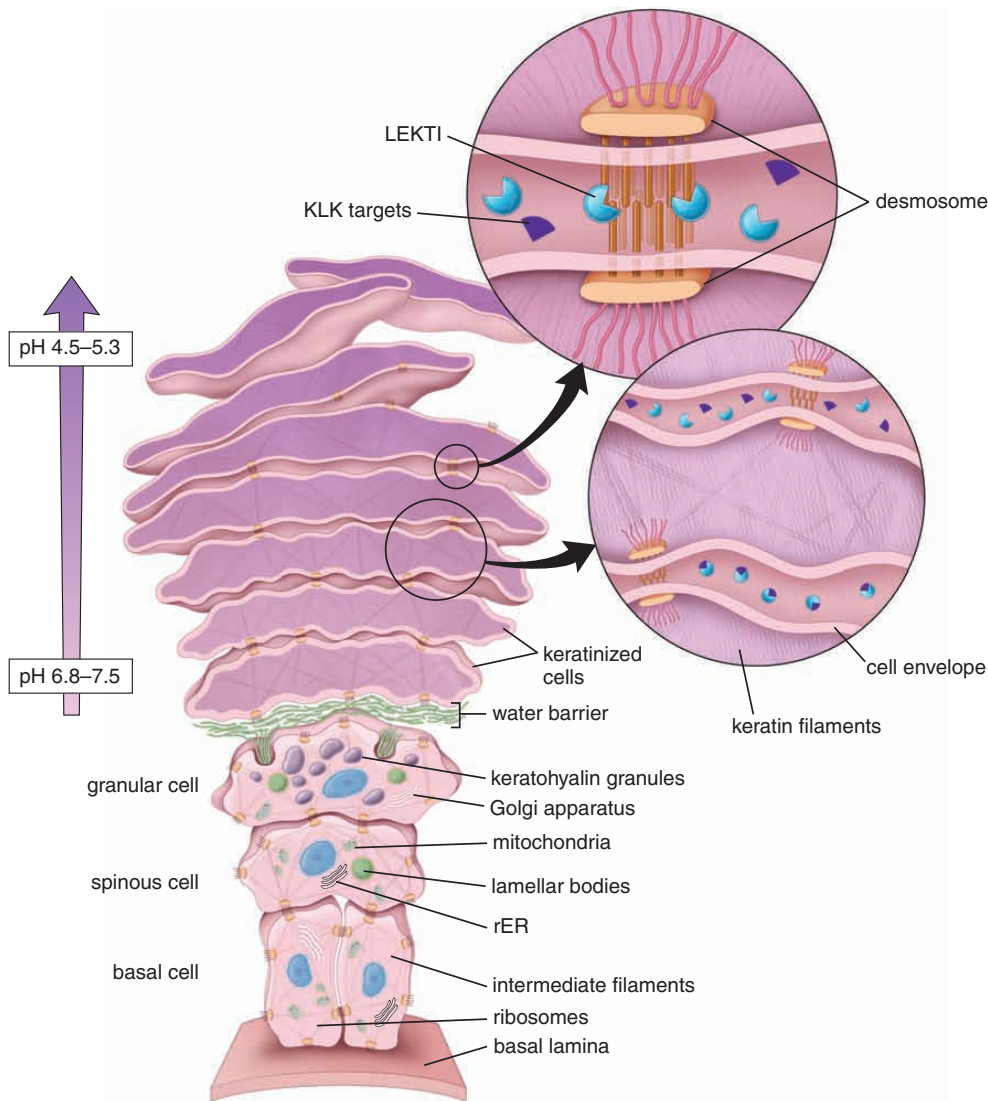


FIGURE 15.4 ▲ Schematic diagram of keratinocytes in the epidermis. The keratinocytes in this figure reflect different stages in the life cycle of the cell as it passes from the basal layer to the skin surface, where it becomes desquamated. The basal cell begins to synthesize intermediate (keratin) filaments; these are grouped into bundles and are seen in the light microscope as tonofibrils. The cell enters the spinous layer, where the synthesis of intermediate filaments continues. In the upper part of the spinous layer, the cells begin to produce keratohyalin granules containing intermediate filament-associated proteins and glycolipid-containing lamellar bodies. Within the granular layer, the cell discharges lamellar bodies that contribute to formation of the water barrier of the epidermis; the remainder of the cell cytoplasm contains numerous keratohyalin granules that, in close association with tonofilaments, form the cell envelope. The surface cells are keratinized; they contain a thick cell envelope and bundles of tonofilaments in a specialized matrix. Desquamation of keratinized cells is controlled by pH that monitors KLK activity and its interaction with LEKTI. Keratinocytes positioned near the granular layer exhibit neutral pH, which maintains desmosomal interactions and allows a strong interaction in the extracellular matrix between LEKTI and its KLK targets. As the pH acidifies toward the skin surface, LEKTI and KLK dissociate, allowing proteinase to activate and search for other protein targets in the extracellular space. In the most superficial layers of keratinocytes, pH is low enough for active KLK molecules to digest desmosomal proteins. In concert with other proteinase activities, this action leads to a complete degradation of desmosomal junctions, resulting in the detachment of the most superficial layer of keratinocytes. *rER*, rough-surfaced endoplasmic reticulum.

to degrade the desmosomes and cause keratinocyte release (see Fig. 15.4). Under normal condition, the process allows a controlled renewal of the epidermis by means of its pH gradient. Recently, pathogenic mutations were identified in the gene called **serine protease inhibitor Kazal-type 5 (SPINK5)**, which encodes LEKTI. **Netherton syndrome**, a rare genetic disorder associated with a defective SPINK5 gene, is characterized by decreased skin barrier function, generalized redness of the skin (erythroderma), and scaling.

Lamellar bodies contribute to the formation of the intercellular epidermal water barrier.

An **epidermal water barrier** is essential for mammalian “dry” epithelia and is responsible for **maintaining body homeostasis**. The barrier is established primarily by two factors in terminally differentiating keratinocytes: (1) deposition of insoluble proteins on the inner surface of the plasma membrane and (2) a lipid layer that is attached to the outer surface of the plasma membrane.

As the keratinocytes in the stratum spinosum begin to produce keratohyalin granules, they also produce membrane-bound **lamellar bodies (membrane-coating granules)**. They are tubular or ovoid-shaped membrane-bound organelles that are unique to mammalian epidermis. Spinous and granular cells synthesize a heterogeneous mixture of **probarrier lipids** and their respective **lipid-processing enzymes** such as glycosphingolipids, phospholipids, ceramides, acidic sphingomyelinase, and secretory phospholipase A²; this mixture is assembled into lamellar bodies in the Golgi apparatus (Fig. 15.5). In addition, lamellar bodies contain **proteases** (i.e., SC chymotryptic enzyme, cathepsin D, acid phosphatase,

glycosidases, protease inhibitors). The contents of the granules are then secreted by exocytosis into the intercellular spaces between the stratum granulosum and stratum corneum. The organization of these intercellular lipid lamellae is responsible for the formation of the **epidermal water barrier** (Fig. 15.6). In addition to their major role in the formation of barrier homeostasis, lamellar bodies are also involved in formation of the cornified envelope, desquamation of cornified cells, and antimicrobial defenses in the skin.

The **epidermal water barrier** thus consists of two structural elements:

- The **cell envelope (CE)** is a 15-nm-thick layer of insoluble proteins deposited on the inner surface of the plasma membrane that contributes to the strong mechanical properties of the barrier. The thickness of the CE increases in epithelia that are subject to considerable mechanical stress (e.g., lip, palm of the hand, sole of the foot). The CE is formed by cross-linking **small proline-rich (SPR) proteins** and larger structural proteins. The structural proteins include **cystatin**, **desmosomal proteins (desmoplakin)**, **elafin**, **envolopakin**, **filaggrin**, **involucrin**, five different **keratin chains**, and **loricrin**. Loricrin is the major structural protein and accounts for almost 80% of the total CE protein mass. This 26 kDa insoluble protein has the highest glycine content of any known protein in the body.
- The **lipid envelope** is a 5-nm-thick layer of lipids attached to the cell surface by ester bonds. The major lipid components of the lipid envelope are **ceramides**, which belong to the class of sphingolipids, **cholesterol**, and **free fatty acids**. However, the most important component is the monomolecular layer of **acylglucosylceramide**, which provides a “Teflon-like” coating on the cell surface. Ceramides also play an important role in cell signaling and are partially responsible for inducing cell differentiation, triggering apoptosis, and reducing cell proliferation. As the cells continue to move toward the free surface, the barrier is constantly maintained by keratinocytes entering the process of terminal differentiation. Lamellae may remain as recognizable discs in the intercellular space or may fuse into broad sheets or layers.

Experiments have shown that the epidermis of animals with induced **essential fatty acid deficiency (EFAD)** is more permeable than normal to water. The membrane-coating granules also have fewer lamellae than normal. Destruction of the **epidermal water barrier** over large areas, as in **severe burns**, can lead to life-threatening loss of fluid from the body.

The epidermis is in a state of dynamic equilibrium in which exfoliated keratinized cells are constantly replaced by a steady flow of terminally differentiated cells.

Epidermal cell replacement is maintained by several processes, including:

- division of basal cells in the stratum basale,
- differentiation and programmed cell death as the cells move upwards to stratum corneum, and
- cell loss by exfoliation from the skin surface.

To maintain this equilibrium, each individual cell in the epidermis has a predetermined amount of time to perform specific functions. Several scientific experiments and

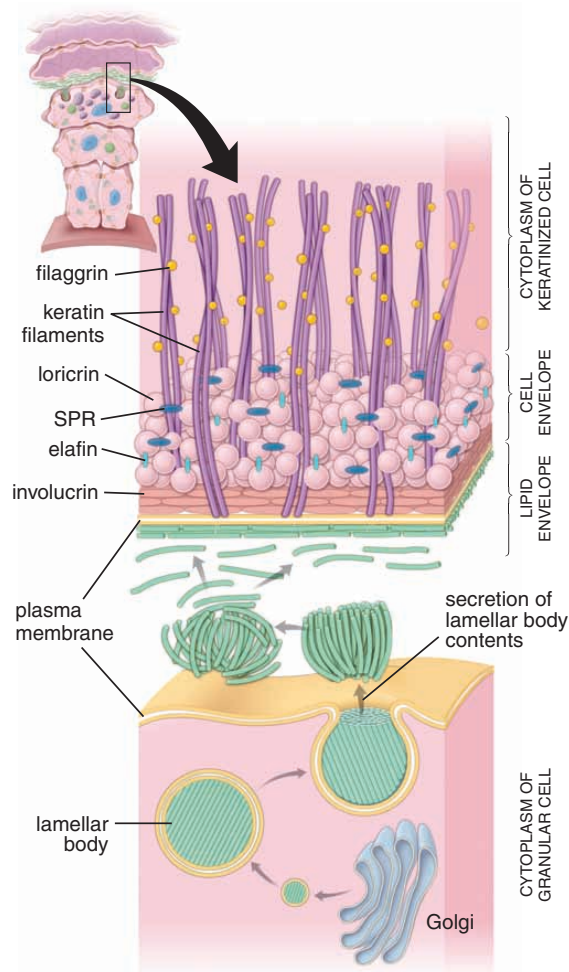


FIGURE 15.5 ▲ Schematic diagram of the epidermal water barrier. The heterogeneous mixture of glycosphingolipids, phospholipids, and ceramides makes up the lamellae of the lamellar bodies. The lamellar bodies, produced within the Golgi apparatus, are secreted by exocytosis into the intercellular spaces between the stratum granulosum and stratum corneum, where they form the lipid envelope. The lamellar arrangement of lipid molecules is depicted in the intercellular space just below the thickened plasma membrane and forms the cell envelope of the keratinized keratinocyte. The innermost part of the cell envelope consists primarily of loricrin molecules (pink spheres) that are cross-linked by small proline-rich (SPR) proteins and elafin. The layer adjacent to the cytoplasmic surface of the plasma membrane consists of the two tightly packed proteins involucrin and cystatin α . Keratin filaments (tonofilaments) bound by filaggrin are anchored into the cell envelope.

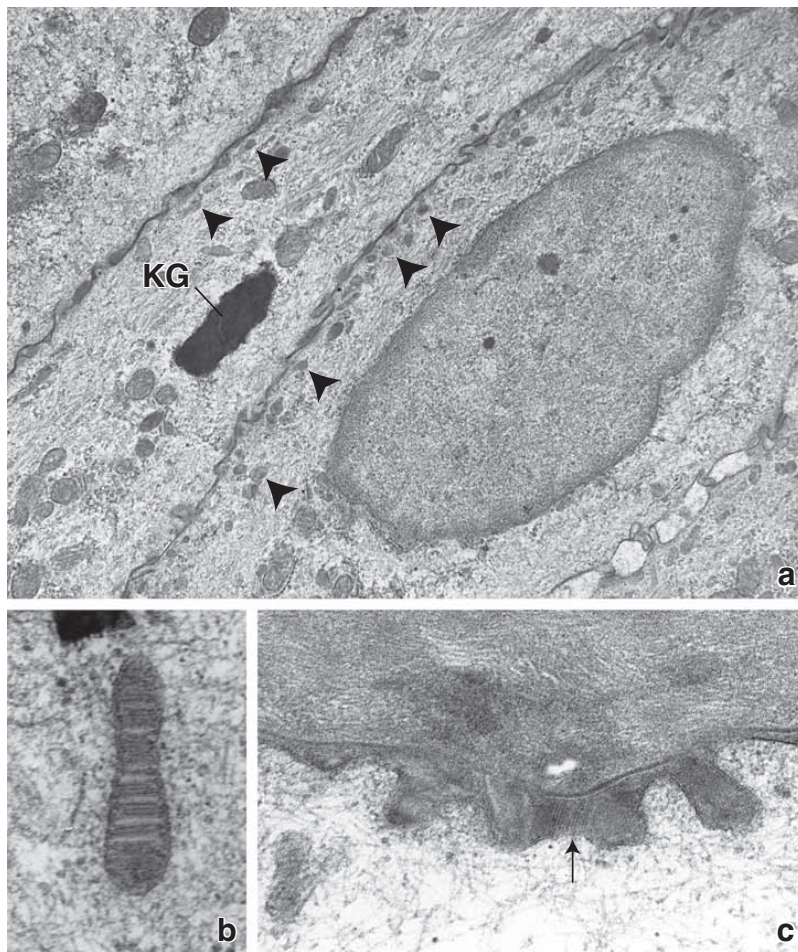


FIGURE 15.6 ▲ **Electron micrographs of keratinocytes.** **a.** Much of the keratinocyte cytoplasm is filled with tonofilaments. One keratinocyte exhibits a keratohyalin granule (KG). Near the plasma membrane closest to the surface (*upper left*), two keratinocytes display lamellar bodies (*arrowheads*). $\times 8,500$. **b.** A lamellar body at higher magnification. $\times 135,000$. **c.** Part of a keratinized cell and the underlying keratinocyte. Located between the cells are the contents of the lamellar bodies, which have been discharged into the intercellular space (*arrow*) to form the lipid envelope. $\times 90,000$. (Courtesy of Dr. Albert I. Farberman.)

empirical calculations concluded that the turnover time for keratinocyte compartment (stratum spinosum and granulosum) is about 31 days, with an additional 14 days for the stratum corneum (average thickness in humans, 16 to 20 cell layers). Adding 1 to 2 days for mitotic divisions in stratum basale, the total epidermal turnover time takes approximately 47 days (Fig. 15.7). One cell layer in the stratum corneum has been found to produce and exfoliate every 22.4 hours. In hyperproliferative diseases, such as **psoriasis**, the epidermal turnover time is faster, taking approximately 8 to 10 days. It is manifested by an increase in epidermal thickness and a decrease in cell death. Clinically, psoriasis appears as elevated, reddish patches of itchy skin, often covered by silvery-white scales. Patches vary in size and typically appear on knees, elbows, lower back, and scalp.

Melanocytes

Neural crest–derived melanocytes are scattered among the basal cells of the stratum basale.

During embryonic life, **melanocyte precursor cells** migrate from the neural crest and enter the developing epidermis. A specific functional association is then established—the **epidermal–melanin unit**—in which one

melanocyte maintains an association with a given number of keratinocytes. In humans, each epidermal–melanin unit is estimated to contain one melanocyte associated with approximately 36 keratinocytes. The ratio of melanocytes to keratinocytes or their precursors in the stratum basale may vary from 1:4 to 1:40 or even higher, depending on the area of the body. This ratio is constant in all races but is influenced by age and environmental factors, such as exposure to sun.

In adults, a pool of undifferentiated melanocyte stem cells resides in the area of the hair follicle called the **follicular bulge**. Differentiation of the melanocyte stem cell is regulated by the expression of the **Pax3 gene** that belongs to the paired box (PAX) family of transcription factors. Pax3 activates expression of the microphthalmia transcription factor (MITF), which is critical for the development and differentiation of melanocytes (melanogenesis). Melanocytes maintain the capacity to replicate throughout their life, although at a much slower rate than keratinocytes, thus maintaining the epidermal–melanin unit.

The epidermal **melanocyte** is a dendritic cell that is scattered among the basal cells of the stratum basale (Fig. 15.8). They are called **dendritic cells** because the rounded cell body resides in the basal layer and extends long processes between the keratinocytes of the stratum spinosum. Neither the

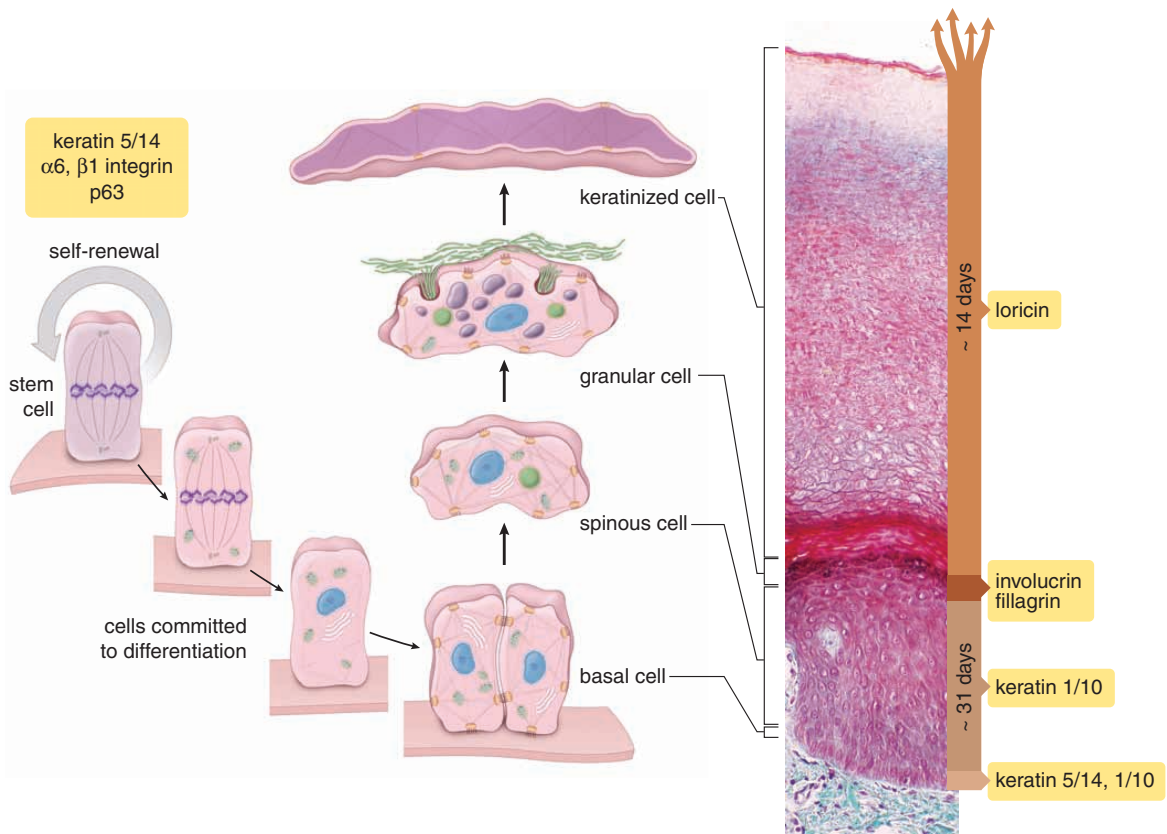


FIGURE 15.7 ▲ **Schematic diagram of epidermal cell differentiation and replacement.** Epidermal cell replacement is initiated by the division of stem cells in the stratum basale. Newly formed cells further undergo divisions in the stratum basale and move upward as they differentiate into keratinized cells, which eventually are lost by exfoliation on the skin surface. To maintain this equilibrium between cell divisions and cell loss at the skin surface, each cell has a predetermined time to travel through specific compartments of the epidermis and to perform specific functions. Mitotic divisions in the stratum basale take about 1 to 2 days; after that, it takes an average 31 days for keratinocytes to move through the stratum spinosum (spinous cells) and differentiate into granular cells in the stratum granulosum. It takes an additional 14 days for the keratinized cell to cross the stratum corneum (assuming an average thickness of 16 to 20 cells in humans). Therefore, the total epidermal turnover time is approximately 47 days. At every stage of differentiation, cells express different molecular markers (see yellow boxes), which can be useful in identifying specific cells using immunocytochemical methods. *Inset on the left* shows full-thickness section of the epidermis from a human fingertip stained with Mallory's trichrome. $\times 260$.

processes nor the cell body forms desmosomal attachments with neighboring keratinocytes. However, melanocytes that reside close to the basal lamina have structures that resemble hemidesmosomes. In routine hematoxylin and eosin (H&E) preparations, melanocytes are seen in the stratum basale with elongated nuclei surrounded by a clear cytoplasm. With the TEM, however, they are readily identified by the developing and mature melanin granules in the cytoplasm (see Fig. 15.8).

Melanocytes produce and distribute melanin into keratinocytes.

The epidermal **melanocytes** produce and secrete the pigment **melanin**. The most important function of melanin is to protect the organism against the damaging effects of nonionizing ultraviolet irradiation. Melanin is produced by the oxidation of **tyrosine** to **3,4-dihydroxyphenylalanine (DOPA)** by **tyrosinase** and the subsequent transformation of DOPA into **melanin**. These reactions initially occur in membrane-bounded, lysosome-related organelles called **premelanosomes**, which are derived from the Golgi apparatus (Fig. 15.9). Melanin synthesis is regulated by the action of melanocyte-stimulating

hormone (MSH). MSH produced by the anterior lobe of the pituitary gland binds to the melanocortin 1 receptor (MC1R) on melanocytes and via G-protein signaling cascade increases the activity of tyrosinase, thus stimulating synthesis of melanin.

Premelanosomes and the **early melanosomes**, which have a low melanin content, exhibit a finely ordered internal structure with the TEM, reflecting their content of tyrosinase molecules. As more melanin is produced by the oxidation of tyrosine, the internal structure of the premelanosome becomes obscured until the mature melanin granule, the **melanosome**, is formed and then appears as an electron-opaque granule. Premelanosomes are concentrated near the Golgi apparatus, nearly mature melanosomes at the bases of the cell processes, and mature melanosomes most commonly in and at the ends of the processes (see Fig. 15.9). Developing melanosomes and their melanin contents are transferred to neighboring keratinocytes by **pigment donation**. This process, which involves the phagocytosis of the tips of the melanocyte processes by keratinocytes, is a type of **cytokine secretion** because a small amount of cytoplasm surrounding the melanosome is also phagocytosed.

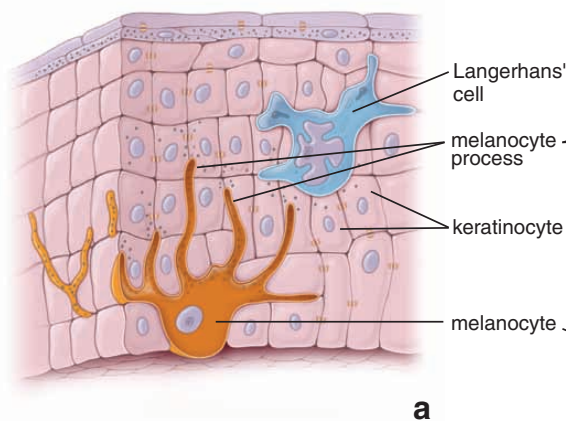
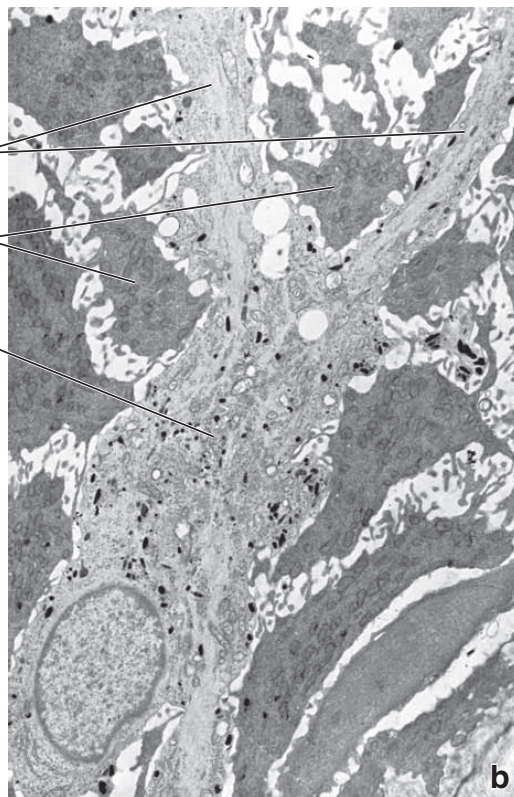
**a****b**

FIGURE 15.8 ▲ **Diagram of the epidermis and electron micrograph of a melanocyte.** **a.** This diagram shows a melanocyte interacting with several cells of the stratum basale and the stratum spinosum. The melanocyte has long dendritic processes that contain accumulated melanosomes and extend between the cells of the epidermis, which are also visible on the electron micrograph. The Langerhans' cell is a dendritic cell often confused with a melanocyte but is actually part of the mononuclear phagocytotic system and functions as an antigen-presenting cell of the immune system in the initiation of cutaneous hypersensitivity reactions (contact allergic dermatitis). **b.** The melanocyte reveals several processes extending between neighboring keratinocytes. The small dark bodies are melanosomes. $\times 8,500$. (Courtesy of Dr. Bryce L. Munger.)

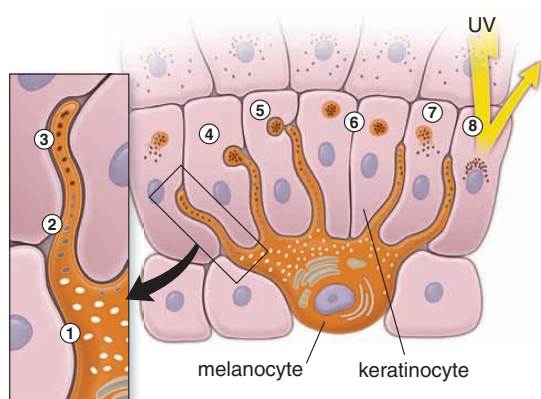


FIGURE 15.9 ▲ **Formation of melanin and mechanism of pigment donation.** Melanocytes produce lysosome-related membrane-bound structures that originate from the Golgi apparatus as pre-melanosomes (1) that are involved in melanin synthesis. Melanin is produced from tyrosine by a series of enzymatic reactions and its accumulation is visible in the early melanosomes (2). As maturation progresses, melanosomes travel toward the ends of the melanocyte processes. Mature melanosomes (3) have a high concentration of melanin and accumulate at the ends of melanocyte processes that invaginate into the keratinocyte's cell membrane (4). Keratinocytes phagocytose the tips of the melanocyte processes containing the melanosomes (5). In the process described as pigment donation, melanin is transferred to neighboring keratinocytes in vesicles containing melanosomes with a small amount of melanocyte cytoplasm (6). Once inside the keratinocytes, melanosomes are released into the cytoplasm (7). Melanosomes are distributed within the keratinocytes with more pronounced accumulation in areas over the nuclei, creating "dark umbrellas" (8) that protect the nuclear DNA from the sun's harmful ultraviolet radiation.

Melanosomes and their content are degraded in the process of macroautophagy with various rates in different individuals. In darker skin individuals, the melanin is degraded slowly and melanosomes remain discrete; in lighter skin, the melanin is degraded more rapidly.

Taking into consideration the complexity of melanin biogenesis, protein trafficking, organelle movement, and cell-to-cell interactions within the epidermal-melanin unit, it is explainable that even minor changes in the cellular environment can affect melanosome structure and the pigment donation process. Numerous intrinsic and extrinsic factors are also responsible for **skin pigmentation**, including age, ethnicity and gender differences, variable hormone levels and affinities of their receptors, genetic defects, ultraviolet radiation, climate and season changes, and chemical exposure to toxins and pollutants.

Langerhans' Cells

Langerhans' cells are antigen-presenting cells in the epidermis.

Langerhans' cells are dendritic-appearing, antigen-presenting cells in the epidermis. They originate from **common lymphoid progenitor (CLP) cells** in bone marrow, migrate via the bloodstream, and ultimately enter the epidermis where they differentiate into immunocompetent cells. Langerhans' cells encounter and process antigens entering through the skin. Therefore, they constitute part of the mononuclear phagocyte system (MPS; page 181) and provide immunosurveillance of the epidermis. Once antigen is phagocytized, processed, and displayed on the surface of the Langerhans' cell, the cell migrates from the epidermis to a regional lymph node where

it interacts with T lymphocytes. Several experimental studies determined that the ratio of Langerhans' cells to other cells in the epidermis of normal human skin is constant at 1:53.

Langerhans' cells cannot be distinguished with certainty in routine H&E-stained paraffin sections. Like melanocytes, Langerhans' cells do not form desmosomes with neighboring keratinocytes. The nucleus stains heavily with hematoxylin, and the cytoplasm is clear. With special techniques, such as gold chloride impregnation or immunostaining with antibody against **CD1a molecules**, Langerhans' cells can be readily seen in the stratum spinosum. They possess dendritic processes resembling those of the melanocyte. The TEM reveals several distinctive features of a Langerhans' cell (Fig. 15.10). Its nucleus is characteristically indented in many places, so the nuclear profile is uneven. Also, it possesses characteristic, tennis racquet-shaped **Birbeck granules**. They represent relatively small vesicles, which appear as rods with a bulbous expansion at their end.

Like macrophages, Langerhans' cells express both MHC I and MHC II molecules, as well as F_c receptors for immunoglobulin G (IgG). Langerhans' cells also express complement C3b receptors as well as fluctuating quantities of CD1a molecules. As an antigen-presenting cell, the **Langerhans' cell** is involved

in **delayed-type hypersensitivity reactions** (e.g., contact allergic dermatitis and other cell-mediated immune responses in the skin) through the uptake of antigen in the skin and its transport to the lymph nodes. Skin biopsy specimens from individuals with **AIDS** or AIDS-related complex reveal that Langerhans' cells contain HIV in their cytoplasm. Langerhans' cells appear to be more resistant than T cells to the deadly effects of HIV and may, therefore, serve as a reservoir for the virus.

In addition, a malignant transformation of Langerhans' cells is responsible for **histiocytosis X** (Langerhans' cell histiocytosis), a group of immune diseases that are characterized by an abnormal increase and spread of Langerhans' cells. Accumulation of these abnormal cells may form tumors, which can affect various parts of the body, including the bones, lungs, skull, and other areas and organs.

Merkel's Cells

Merkel's cells are epidermal cells that function in cutaneous sensation.

Merkel's cells are dendritic cells located in the stratum basale. The origin of Merkel's cells is unknown; they possess

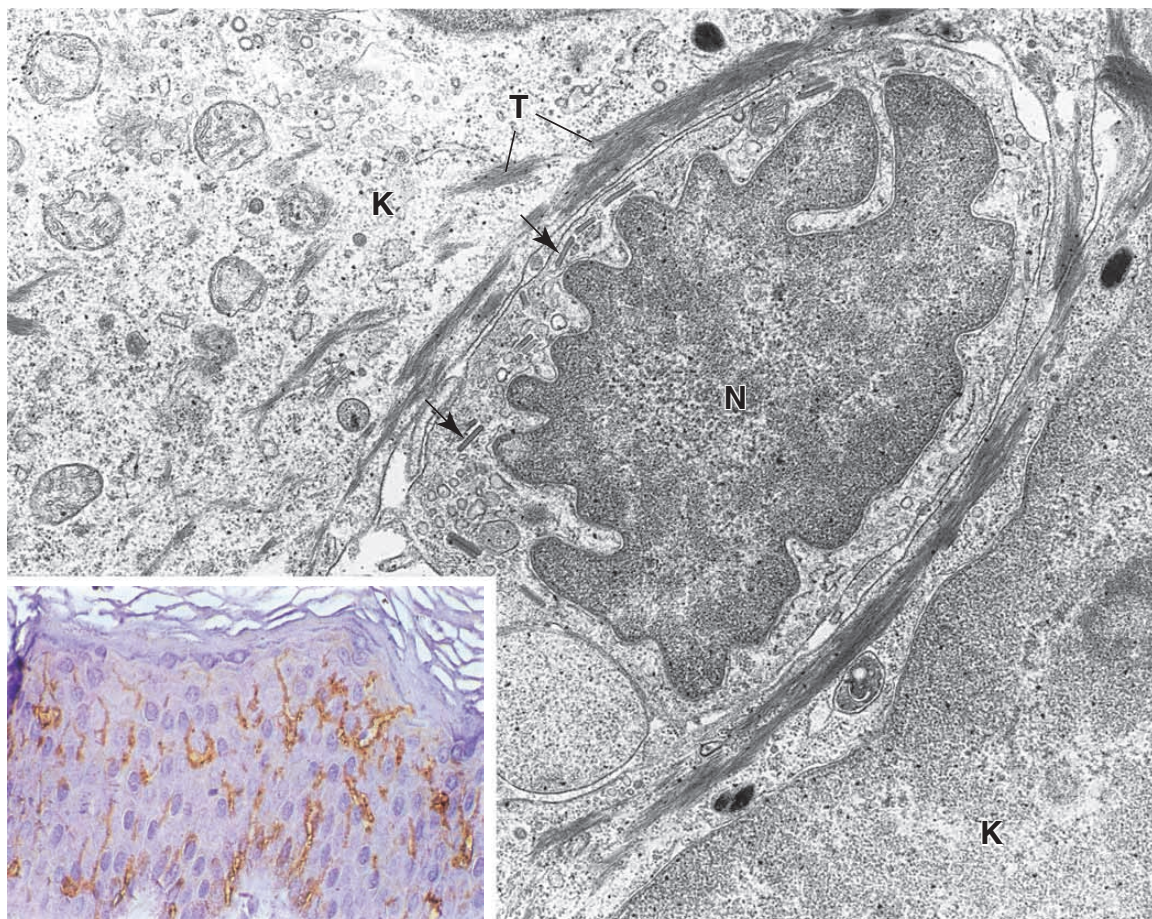


FIGURE 15.10 ▲ Electron micrograph of a Langerhans' cell. The nucleus (N) of a Langerhans' cell is characteristically indented in many places, and the cytoplasm contains distinctive rod-shaped bodies (arrows). Note the presence of tonofilaments (T) in adjacent keratinocytes (K) but the absence of these filaments in the Langerhans' cell. $\times 19,000$. **Inset.** Photomicrograph of the epidermis shows the distribution and dendritic nature of the Langerhans' cells that were stained via immunostaining techniques with antibodies against CD1a surface antigen. $\times 300$. (Reprinted with permission from Urmacher CD. Normal Skin. In: Sternberg SS, ed. *Histology for Pathologists*. Philadelphia: Lippincott-Raven, 1997:25–45.)

antigenic markers of both epidermal and neural type. They are most abundant in skin where sensory perception is acute, such as the fingertips. Merkel's cells are bound to adjoining keratinocytes by desmosomes and contain intermediate (keratin) filaments in their cytoplasm. The nucleus is lobed, and the cytoplasm is somewhat denser than that of melanocytes and Langerhans' cells. They may contain some melanosomes in their cytoplasm, but they are best characterized by the presence of **80-nm dense-cored neurosecretory granules** that resemble those found in the adrenal medulla and carotid body (Fig. 15.11). Merkel's cells are closely associated with the expanded terminal bulb of afferent myelinated nerve fibers. The neuron terminal loses its Schwann cell covering and immediately penetrates the basal lamina, where it expands into a plate-like ending called a *disc receptor* that lies in close apposition to the base of the Merkel's cell. The combination of the neuron and epidermal cell, called a **Merkel's corpuscle**, is a sensitive **mechanoreceptor**.

Merkel cell carcinoma (MCC) is a rare but highly aggressive type of skin cancer that develops when Merkel cells undergo uncontrolled proliferation. It starts most often in areas of skin exposed to the sun, such as the head, neck, and upper and lower limbs. MCC tends to grow quickly and to metastasize via lymph vessels at an early stage.



STRUCTURES OF SKIN

Nerve Supply

The skin is endowed with sensory receptors of various types that are peripheral terminals of sensory nerves (Fig. 15.12). It is also well supplied with motor nerve endings to the blood vessels, arrector pili muscles, and sweat glands.

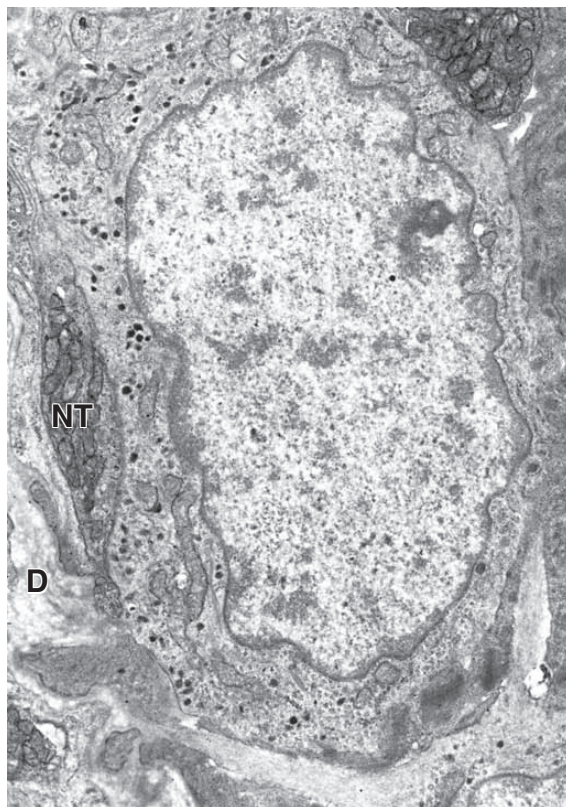


FIGURE 15.11 ▲ **Electron micrograph of a Merkel's cell.** The cell has small neurosecretory granules in the cytoplasm and makes contact with a peripheral nerve terminal (NT) of a sensory neuron. The dermis (D) is in the lower part of the micrograph. $\times 14,450$. (Courtesy of Dr. Bryce L. Munger.)

FOLDER 15.2 Functional Considerations: Skin Color

The **color of an individual's skin** is caused by several factors, including major genetic determinants, several modifier genes, and environmental influences such as exposure to ultraviolet radiation and gender effects. The most significant is **melanin content**. Although the number of melanocytes is essentially the same in all races, the fate of the melanin that is produced by the melanocytes differs. For example, because of the lysosomal activity of the keratinocytes, melanin is degraded more rapidly in individuals with light skin than in individuals with dark skin. In the former, melanosomes are more concentrated in the keratinocytes nearest the basal layer and are relatively sparse in the midregion of the stratum granulosum. In contrast, dark skin may exhibit melanosomes throughout the epidermis, including the stratum corneum.

In addition, melanin pigment comprises two distinct forms. One form, **eumelanin**, is a brownish black pigment. The other form, **pheomelanin**, is a reddish yellow pigment. Each is genetically determined. Coloration is most apparent in hair because of the concentration of melanin pigment granules, but it is also reflected in skin coloration.

Exposure to **ultraviolet radiation**, particularly the sun's rays, is called **tanning**. It increases the number of

melanocytes and accelerates the rate of melanin production, thus protecting against further radiation effects. Response to ultraviolet radiation is genetically determined and is more pronounced in individuals with darker color of skin.

Increased pigmentation of the skin may also result from hormonal imbalance—for example, in Addison's disease. Lack of pigmentation occurs in a condition known as **albinism**. In this hereditary condition, premelanosomes are produced by melanocytes, but because of the absence of tyrosinase, the transformation of tyrosine into 3,4-dihydroxyphenylalanine (DOPA) and the subsequent transformation of DOPA into melanin fail to occur. Thus, there is no pigmentation in the skin or hair of these individuals.

Two genes—*Bcl2* and *Mitf*—appear to be responsible for the **process of graying**. The expression of *Bcl2* in melanocyte stem cells is essential to maintaining their population within the niche of the follicular bulge. Deficiency in *Bcl2* expression causes apoptosis of melanocyte stem cells and a consequent decrease in the number of melanocytes. Melanocyte depletion occurs with age, resulting in a decreased rate of pigment donation to keratinocytes. Therefore, the skin becomes lighter with increased age, and the incidence of skin cancer also increases. Melanocyte

depletion caused by defective self-maintenance of melanocyte stem cells is also linked to hair graying, the most obvious sign of aging in humans. Individuals with a mutation in the *Bcl2* gene may become prematurely gray.

Other normal factors that affect skin coloration include the presence of oxyhemoglobin in the dermal vascular bed, which imparts a red hue; the presence of carotenes, an exogenous orange pigment taken up from foods and concentrated in tissues containing fat; and the presence

of certain endogenous pigments. The latter include degradation products of hemoglobin, iron-containing hemosiderin, and iron-free bilirubin, all of which impart color to the skin. Hemosiderin is a golden brown pigment, whereas bilirubin is a yellowish brown pigment. Bilirubin is normally removed from the bloodstream by the liver and eliminated via the bile. A yellowish skin color as a result of abnormal accumulation of bilirubin reflects liver dysfunction and is evidenced as **jaundice**.

Free nerve endings are the most numerous neuronal receptors in the epidermis.

Free nerve endings in the epidermis terminate in the stratum granulosum. The endings are “free” in that they lack a connective tissue or Schwann cell investment. Such neuronal endings subserve multiple sensory modalities including fine touch, heat, cold, and pain without apparent morphologic distinction. Networks

of free dermal endings surround most hair follicles and attach to their external root sheath (Figs. 15.13 and 15.14). In this position, they are particularly sensitive to hair movement and serve as mechanoreceptors. This relationship imparts a sophisticated degree of specialization in the receptors that surround tactile hairs (vibrissae), such as the whiskers of a cat or rodent, in which each vibrissa has a specific representation in the cerebral cortex.

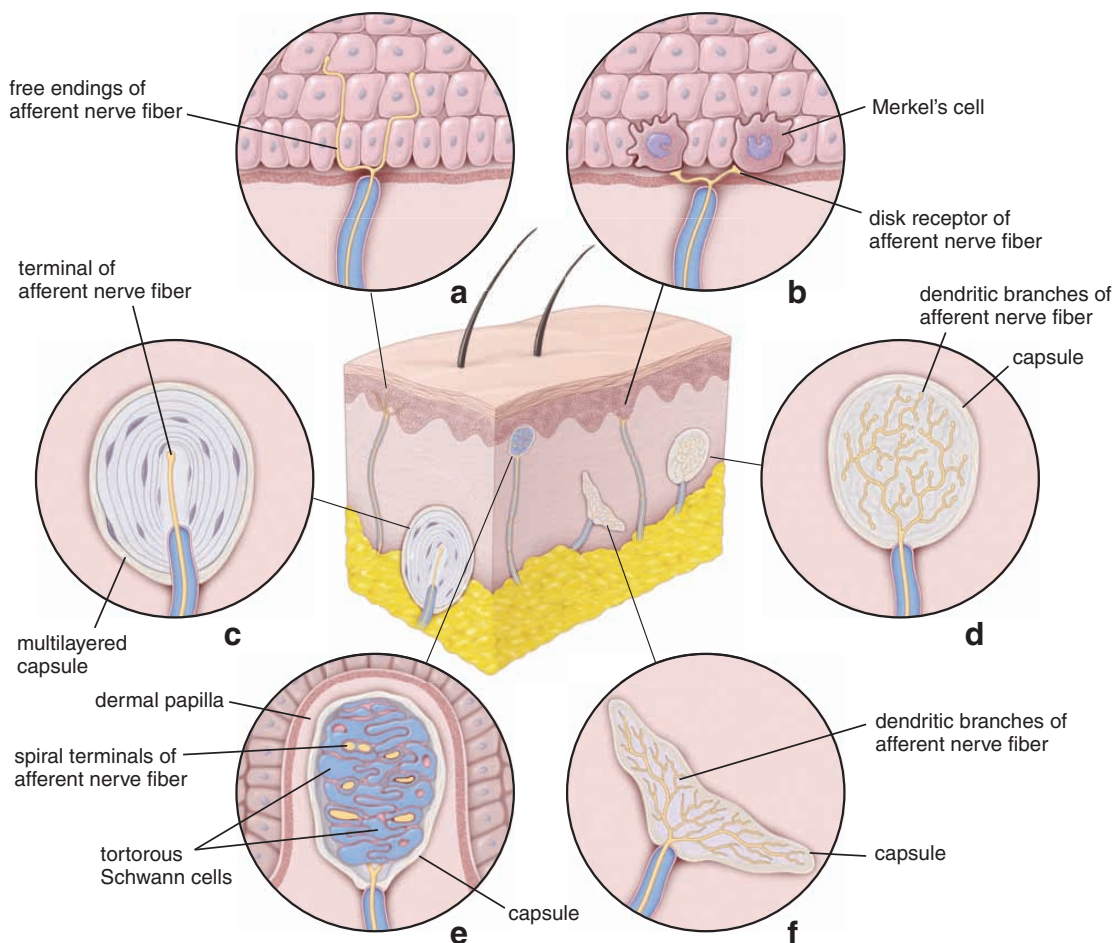


FIGURE 15.12 ▲ **Diagram of the sensory receptors in skin.** **a.** Epidermal free endings. **b.** Merkel's corpuscles containing Merkel's cells and disc receptors of afferent myelinated nerve fiber. **c.** Pacinian corpuscle located in the deep layer of deep dermis and hypodermis. **d.** Krause's end bulb serves as cold receptor. **e.** Meissner's corpuscle in dermal papilla. **f.** Ruffini's corpuscle in deep layers of the dermis. Note that sensory nerve fibers in receptors **c-f** are encapsulated.

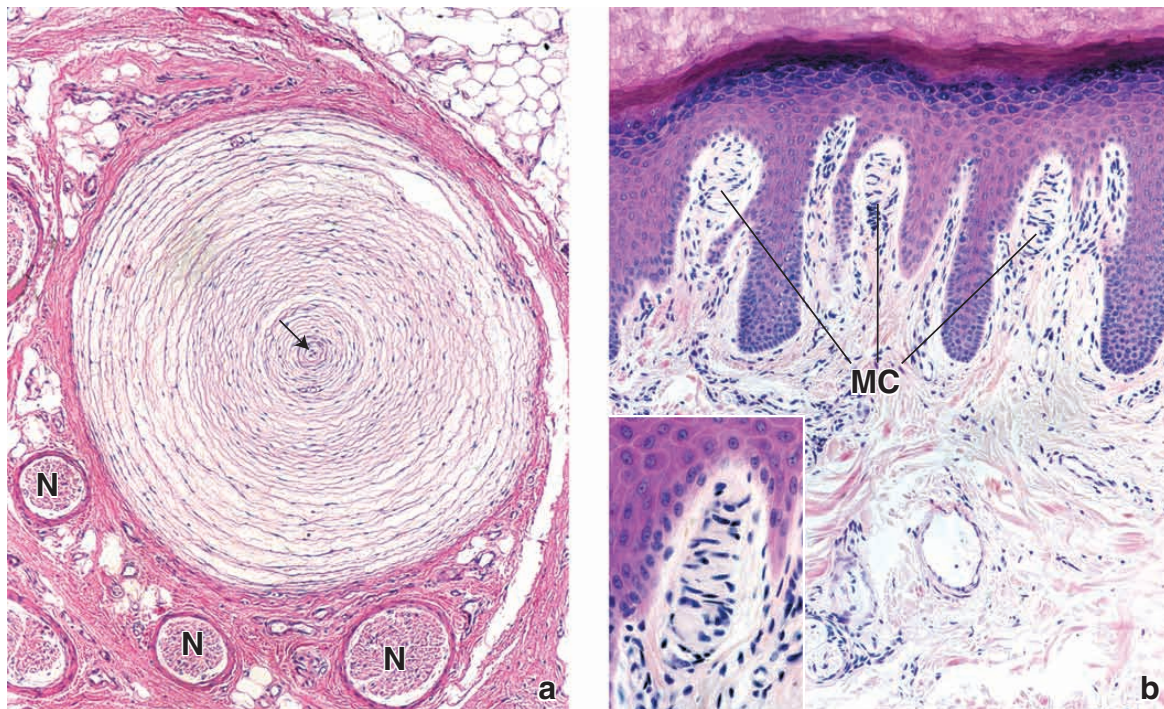


FIGURE 15.13 ▲ Pacinian and Meissner's corpuscles in H&E preparations. a. In this photomicrograph, the concentric cellular lamellae of the Pacinian corpuscle are visible because of flat, fibroblast-like supportive cells. Although not evident within the tissue section, these cells are continuous with the endoneurium of the nerve fiber. The spaces between lamellae contain mostly fluid. The neural portion of the Pacinian corpuscle travels longitudinally through the center of the structure (*arrow*). Several nerves (*N*) are present adjacent to the corpuscle. $\times 85$. **b.** Three Meissner's corpuscles (*MC*) are shown residing within the dermal papillae. Note the direct proximity of the corpuscle to the undersurface of the epidermis. $\times 150$. **Inset.** A higher magnification of a Meissner's corpuscle. The nerve fiber terminates at the superficial pole of the corpuscle. Note that supporting cells are oriented approximately at right angles to the long axis of the corpuscle. $\times 320$.

Other nerve endings in the skin are enclosed in a connective tissue capsule. **Encapsulated nerve endings** include the following:

- **Pacinian corpuscles** detect pressure changes and vibrations applied on the skin surface.
- **Meissner's corpuscles** are responsible for sensitivity to light touch.
- **Ruffini's corpuscles** are sensitive to skin stretch and torque.

Pacinian corpuscles are deep pressure receptors for mechanical and vibratory pressure.

Pacinian corpuscles are large ovoid structures found in the deeper dermis and hypodermis (especially in the fingertips), in connective tissue in general, and in association with joints, periosteum, and internal organs. Pacinian corpuscles usually have macroscopic dimensions, measuring more than 1 mm along their long axis. They are composed of a myelinated nerve ending surrounded by a capsule structure (see Figs. 15.12 and 15.13a). The nerve enters the capsule at one pole with its myelin sheath intact. The myelin is retained for one or two nodes and is then lost. The unmyelinated portion of the axon extends toward the opposite pole from which it entered, and its length is covered by a series of tightly packed, flattened Schwann cell lamellae that form the inner core of the corpuscle. The remainder or bulk of the capsule, the outer core, is formed by a series of concentric lamellae; each lamella is separated from its neighbor by a narrow

space containing lymph-like fluid (Plate 46, page 522). The appearance of the concentric lamellae as observed in the light microscope is reminiscent of the cut surface of a hemisected onion. Each lamella is composed of flattened cells that correspond to the cells of the endoneurium outside the capsule. In addition to fluid between the lamellae, collagen fibrils are present, although sparse, as are occasional capillaries.

Pacinian corpuscles respond to **pressure** and **vibration** through the displacement of the capsule lamellae. This displacement effectively causes depolarization of the axon.

Meissner's corpuscles are localized within dermal papillae and serve as touch receptors.

Meissner's corpuscles (see Figs. 15.12 and 15.13b) are touch receptors that are particularly responsive to **low-frequency stimuli** in the papillary layer of hairless skin (e.g., the lips and the palmar and volar surfaces, particularly those of the fingers and toes). Generally, they are tapered cylinders that measure about 150 μm along their long axis and are oriented perpendicular to the skin surface. Meissner's corpuscles are present in the dermal papillae just beneath the epidermal basal lamina (Plate 46, page 522). Within these receptors, one or two unmyelinated endings of myelinated nerve fibers follow spiral paths in the corpuscle. The cellular component consists of flattened Schwann cells that form several irregular lamellae through which the axons course to the pole of the corpuscle. In H&E-stained slides of sagittal sections, this structure resembles a loose, twisted skein of wool. It is the Schwann cells that give this impression.

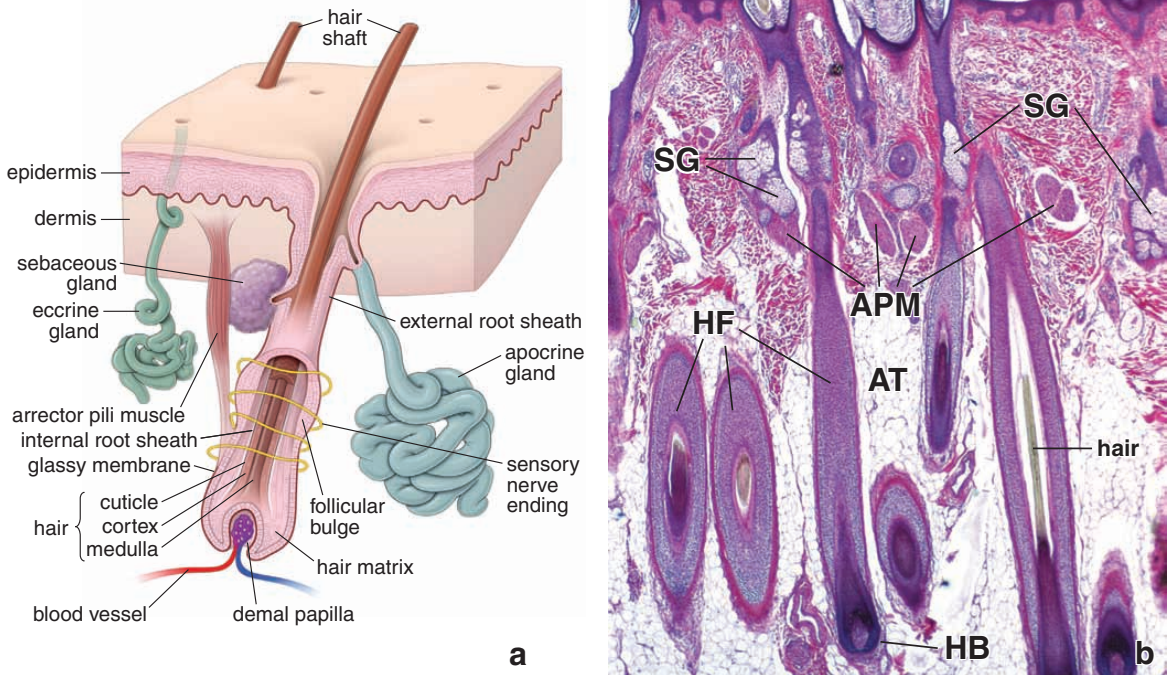


FIGURE 15.14 ▲ Hair follicle and other skin appendages. **a.** Diagram showing a hair follicle. Note the cell layers that form the hair shaft and the surrounding external and internal root sheaths. The sebaceous gland consists of the secretory portion and a short duct that empties into the infundibulum, the upper part of the hair follicle. The arrector pili muscle accompanies the sebaceous gland; contraction of this smooth muscle assists in gland secretion and discharges the sebum into the infundibulum of the hair follicle. Projection of the external root sheath near insertion of the arrector pili muscle forms the follicular bulge that contains epidermal stem cells. Nerve endings (yellow) surround the follicular bulge with nearby insertion of arrector pili muscle. The apocrine sweat gland also empties into the infundibulum. Note that eccrine sweat glands are independent structures and are not associated directly with the hair follicle. **b.** Photomicrograph of H&E-stained section of thin skin from human scalp. The growing end of a hair follicle consists of an expanded hair bulb (HB) of epithelial cells that is invaginated by a connective tissue dermal papilla. Hair matrix that fills the bulb consists of cells that differentiate into the hair shaft and the internal root sheath of the hair follicle (HF). Note that several oblique and longitudinal sections of the hair follicles are embedded in the adipose tissue (AT) of the hypodermis. Some of them reveal a section of the hair. Sebaceous glands (SG) are visible in conjunction with the infundibulum of the hair follicle. $\times 60$. APM, arrector pili muscle.

Ruffini's corpuscles respond to mechanical displacement of adjacent collagen fibers.

Ruffini's corpuscles are the simplest encapsulated mechanoreceptors. They have an elongated fusiform shape and measure 1 to 2 μm in length (see Fig. 15.12f). Structurally, they consist of a thin connective tissue capsule that encloses a fluid-filled space. Collagen fibers from the surrounding connective tissue pass through the capsule. The neural element consists of a single myelinated fiber that enters the capsule, where it loses its myelin sheath and branches to form a dense arborization of fine axonal endings, each terminating in a small knob-like bulb. The axonal endings are dispersed and intertwined inside the capsule. The axonal endings respond to displacement of the collagen fibers induced by sustained or continuous mechanical stress; thus, they respond to **stretch** and **torque**. Ruffini's corpuscles functionally belong to the family of rapidly adapting receptors (phasic receptors) that generate short action potentials at the beginning and end of a stimulus.

Epidermal Skin Appendages

Skin appendages are derived from downgrowths of epidermal epithelium during development. They include the following:

- **Hair follicles** and their product, **hair**
- **Sebaceous glands** and their product, **sebum**

- **Eccrine sweat glands** and their product, **sweat**
- **Apocrine sweat glands** and their mixed product containing a form of sweat with a high concentration of carbohydrates, lipids, and proteins

Both hairs and sweat glands play specific roles in the regulation of body temperature. Sebaceous glands secrete an oily substance that may have protective functions. Apocrine glands produce a serous secretion containing pheromones that act as a sex attractant in other animals and possibly in humans. The epithelium of the skin appendages (especially hair follicles) can serve as a source of new epithelial stem cells for skin wound repair.

Hair Follicles and Hair

Each hair follicle represents an invagination of the epidermis in which a hair is formed.

Hair follicles and **hairs** are present over almost the entire body; they are absent only from the sides and palmar surfaces of the hands, sides and plantar surfaces of the feet, the lips, and the region around the urogenital orifices. Hair distribution is influenced to a considerable degree by sex hormones; these include, in the male, the thick, pigmented facial hairs that begin to grow at puberty and the pubic and axillary hair that develops at puberty in both genders. In the male, the

FOLDER 15.3 Functional Considerations: Hair Growth and Hair Characteristics

Unlike the renewal of the surface epidermis, hair growth is not continuous but a cyclic process. A **period of growth (anagen)** in which a new hair develops is followed by a brief period in which **growth stops (catagen)**. Catagen is followed by a long **rest period (telogen)** in which the follicle atrophies, and the hair is eventually lost. Epidermal stem cells found in the follicular bulge are capable of providing stem cells that give rise to mature anagen follicles. During the hair growth cycle, mature anagen hairs periodically undergo apoptosis and regress to the catagen stage. In this phase, entire follicles retract toward the epidermal layer. As the base of the retracted follicle approximates the follicular bulge, the hair shaft is no longer supported by the nutrient-rich anagen bulb and eventually is ejected from the resting telogen follicle. This makes room for a new shaft that will grow during anagen regeneration.

More than 80% of the hair present in the normal scalp is in the anagen phase. In catagen, the germinative zone is reduced to an epithelial strand still attached to a remnant of the dermal papilla. In the telogen phase, the atrophied

follicle may contract to one half or less of its original length. The hair may remain attached to the follicle for several months during this stage and is called a **club hair** because of the shape of its proximal end.

Hairs vary in size from long, coarse **terminal hairs** that may reach a meter or more in length (scalp hair and beard hair in males) to short, fine **vellus hairs** that may be visible only with the aid of a magnifying lens (vellus hairs of the forehead and anterior surface of the forearm). Terminal hairs are produced by large-diameter, long follicles; vellus hairs are produced by relatively small follicles. Terminal hair follicles may spend up to several years in anagen and only a few months in telogen. In the balding individual, large terminal follicles are gradually converted into small vellus follicles after several growth cycles. The ratio of vellus follicles to terminal follicles increases as baldness progresses. The “completely bald” scalp is not hairless but is populated by vellus follicles that produce fine hairs and remain in telogen for relatively long periods.

hairline tends to recede with age; in both genders, the scalp hair thins with age because of reduced secretion of estrogen and estrogen-like hormones.

The **hair follicle** is responsible for the production and growth of a hair. Coloration of the hair is attributable to the content and type of melanin that the hair contains. The follicle varies in histologic appearance, depending on whether it is in a growing or a resting phase. The growing follicle shows the most elaborate structure; thus, it is described here.

The hair follicle is divided into four regions:

- The **infundibulum** extends from the surface opening of the follicle to the level of the opening of its sebaceous gland. The infundibulum is a part of the **pilosebaceous canal**, which is used as a route for the discharge of the oily substance **sebum**.
- The **isthmus** extends from the infundibulum to the level of insertion of the arrector pili muscle.
- The **follicular bulge** protrudes from the hair follicle near the insertion of the arrector pili muscle and contains **epidermal stem cells** (see Fig. 15.14).
- The **inferior segment** in the growing follicle (see Fig. 15.14) is of nearly uniform diameter except at its base, where it expands to form the **bulb**. The base of the bulb is invaginated by a tuft of vascularized loose connective tissue called, not surprisingly, a **dermal papilla** (Plate 47, page 524).

Other cells forming the bulb, including those that surround the connective tissue dermal papilla, are collectively referred to as the **hair matrix**, which consists simply of **matrix cells**. Matrix cells immediately adjacent to the dermal papilla represent the population of rapidly dividing and differentiating cells that migrated from the follicular bulge containing epidermal stem cells (Fig. 15.15). Division and proliferation of these cells accounts for the growth of the hair. Scattered melanocytes are also present in this germinative layer. They contribute

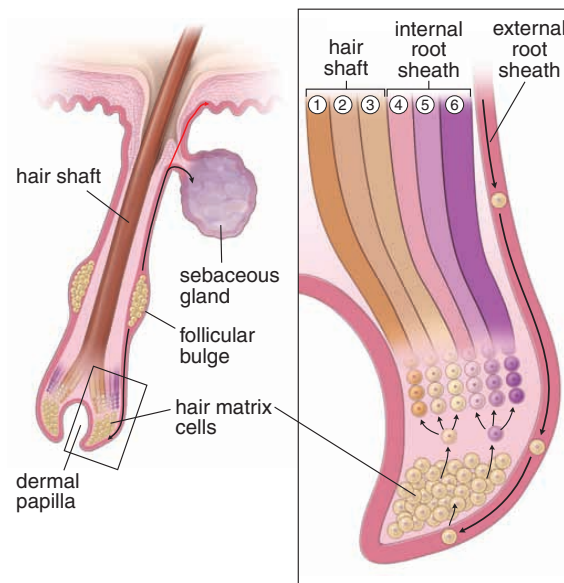


FIGURE 15.15 ▲ Hair follicle and pathways of epidermal stem cell migration. This diagram shows the location and migration pathways of epidermal stem cells that reside in the follicular bulge. Under normal conditions, epidermal stem cells migrate upward to the sebaceous gland and downward to reach the hair matrix in the bulb of the follicle (black arrows). Hair matrix is formed by differentiating cells that migrate along the external root sheath from the follicular bulge. As the differentiation progresses, cells leave the matrix; they form cell layers that differentiate into the hair shaft containing (1) medulla, (2) hair cortex, and (3) hair cuticle, and the internal root sheath containing (4) internal root sheath cuticle, (5) Huxley's layer, and (6) Henle's layer. During injury of the epidermis, the epidermal stem cells migrate from the follicular bulge toward the skin surface (red arrow) and participate in the initial resurfacing of damaged epidermis.

The role of **sebum** is not clearly defined. Various investigators have ascribed bacteriostatic, emollient, barrier, and pheromone functions to sebum. Sebum does appear to play a critical role in the development of acne. The amount of sebum secreted increases significantly at puberty in both males and females. Triglycerides contained in sebum are broken down to fatty acids

by bacteria on the skin surface, and the free fatty acids liberated may be an irritant in the formation of **acne lesions**. On histologic examination, acne is characterized by retention of the sebum in the isthmus of the hair follicle, with variable lymphocytic infiltration. In severe cases, dermal abscesses may form in association with inflamed hair follicles.

melanosomes to the developing hair cells in a manner analogous to that in the stratum basale of the epidermis. The dividing matrix cells differentiate into the keratin-producing cells of the hair and the **internal root sheath**.

The internal root sheath is a multilayered cellular covering that surrounds the deep part of the hair. The internal root sheath has three layers:

- **Henle's layer** consists of an outer single layer of cuboidal cells. These cells are in direct contact with the outermost part of the hair follicle, which represents a downgrowth of the epidermis and is designated the **external root sheath**.
- **Huxley's layer** consists of a single or double layer of flattened cells that form the **middle plate of the internal root sheath**.
- The **internal root sheath cuticle** consists of squamous cells whose outer free surface faces the hair shaft.

A niche of epidermal stem cells that resides within the follicular bulge of the external root sheath provides stem cells for hair growth and skin regeneration.

The progression of the external root sheath of the hair follicle upward toward the epidermal surface reveals the insertion site of the arrector pili muscle and the origin of the sebaceous duct and gland from the wall of the follicular canal (see Fig. 15.14). Nerve endings surround the external root sheath at the level of arrector pili muscle insertion. In this general region resides an aggregate of relatively undifferentiated epithelial cells called the **follicular bulge**. Recent studies identified the follicular bulge as a **niche of epidermal stem (ES) cells** (see Fig. 15.15). The ES cells can reside in this area indefinitely and undergo self-renewal or differentiation into specific cell lineages. Under normal conditions, ES cells are responsible for providing stem cells for the growth of hair follicles (the hair matrix, internal

root sheath, cortex, and medulla) as well as sebaceous glands (see Fig. 15.15). The ES cells that normally reside in the follicular bulge do not contribute to the population of the basal stem cells of the epidermis. However, when the epidermis is injured or lost (such as in extensive skin burns and superficial skin wounds), the ES cells become reprogrammed, migrate toward the wound surface from their follicular niches, and participate in the initial resurfacing of the wound.

Hairs are composed of keratinized cells that develop from hair follicles.

Keratinization of the hair and internal root sheath occurs shortly after the cells leave the matrix in a region called the **keratogenous zone** in the lower third of the follicle. As the cortical cells pass through this zone, they differentiate, extrude their organelles, and become tightly packed with cross-linked keratin intermediate filaments. By the time the hair emerges from the follicle, it is entirely keratinized as **hard keratin**. The internal root sheath, consisting of soft keratin, does not emerge from the follicle with the hair but is broken down at about the isthmus level where sebaceous secretions enter the follicle. A thick basal lamina, called the **glassy membrane**, separates the hair follicle from the dermis. Surrounding the follicle is a dense irregular connective tissue. The **arrector pili muscle** is attached near the follicular bulge, which, as was indicated earlier, serves as an epidermal stem cell niche.

Hairs are elongated filamentous structures that project from the hair follicles. They are composed of heavily cross-linked hard keratins and consist of three layers (see Fig. 15.14):

- The **medulla** forms the central part of the shaft and contains a column of large, loosely connected keratinized cells containing soft keratin. The medulla is present only in thick hairs.

FOLDER 15.5 Clinical Correlation: Sweating and Disease

Although many neural and emotional factors can alter the **composition of sweat**, altered sweat composition can also be a sign of disease. For example, elevated sodium and chloride levels in sweat can serve as an indicator of **cystic fibrosis**. Individuals with cystic fibrosis have two to five times higher than normal amounts of sodium and chloride in their sweat.

In pronounced **uremia**, when the kidneys are unable to rid the body of urea, the concentration of urea in sweat increases. In this condition, after the water evaporates, crystals may be discerned on the skin, especially on the upper lip. These include urea crystals and are called **urea frost**.

- The **cortex** is the largest layer and accounts for approximately 80% of total hair mass. It is located outside the medulla and is composed of cortical cells filled with hard keratin intermediate filaments. Each filament is surrounded by an amorphous space containing **keratin-associated proteins (KAPs)**. These high-sulfur KAPs are responsible for forming the rigid hair shaft by extensive cross-linking of keratin intermediate filaments by disulfide bonds. The cortex determines texture, elasticity, and the color of hair. Melanin pigment responsible for the color of hair is produced by melanocytes present in the germinative layer of the hair bulb.
- The **cuticle of the hair shaft** is the outermost layer of the hair. It contains several layers of overlapping, semitransparent keratinized squamous cells. These cells resemble fish scales or roof tiles with their free edges lying away from the hair follicle. The cuticle protects the hair from physical and chemical damage and determines its porosity.

The **hair structure** in humans is determined by multiple genetic factors that include 17 keratin genes (11 genes for type I and 6 genes for type II keratins) and more than 85 KAP genes.

Sebaceous Glands

Sebaceous glands secrete sebum that coats the hair and skin surface.

Sebaceous glands develop as outgrowths of the external root sheath of the hair follicle, usually producing several glands per follicle (Fig. 15.16 and Plate 45, page 520). The oily substance produced in the gland, **sebum**, is the product of holocrine secretion. The entire cell produces and becomes filled with the fatty product while it simultaneously undergoes programmed cell death (apoptosis) as the product fills the cell. Ultimately, both the secretory product and cell debris are discharged from the gland as sebum into the infundibulum of a hair follicle, which forms the **pilosebaceous canal** with the short duct of the sebaceous gland. New cells are produced by mitosis of the basal cells at the periphery of the gland, and the cells of the gland remain linked to one another by desmosomes. The basal lamina of these cells is continuous with that of the epidermis and the hair follicle. The process of sebum production from the time of basal cell mitosis to the secretion of the sebum takes about 8 days.

The basal cells of the sebaceous gland contain smooth-surface endoplasmic reticulum (sER), rER, free ribosomes, mitochondria, glycogen, and a well-developed Golgi apparatus. As the cells move away from the basal layer and begin to produce the lipid secretory product, the amount of sER increases, reflecting the role of the sER in lipid synthesis and secretion. The cells gradually become filled with numerous lipid droplets separated by thin strands of cytoplasm.

Sweat Glands

In general, **sweat glands** are classified on the bases of their structure and the nature of their secretion. Two types of sweat glands are recognized:

- **Eccrine sweat glands** are distributed over the entire body surface except for the lips and part of the external genitalia.
- **Apocrine sweat glands** are limited to the axilla, areola, and nipple of the mammary gland; skin around the anus;



FIGURE 15.16 ▲ Photomicrograph of a sebaceous gland. a.

This micrograph shows the secretory lobules and their pilosebaceous canal of two sebaceous glands. The canal of the gland on the **left** is about to enter the hair follicle seen at the **top** of the micrograph. The canal of the sebaceous gland on the **right** has been sectioned in a manner that shows mostly the wall of the canal. **×60. b.** The secretory component of the lobule in the **lower box of a** is shown here at higher magnification. Note the light staining of the secretory cells due to the lack of staining of the sebum that they contain. These cells are actively producing sebum. The basal cells at the periphery of the lobule divide and replenish the population of new sebum-producing cells. Also, the arrector pili muscle (APM) composed of smooth muscle fibers is visible in the periphery of the secretory lobule. **×120. c.** The secretory component of the lobule in the **upper box of a** is shown here at higher magnification. The sebum-filled cells are now within the canal. Note their pyknotic nuclei, signifying the death of the cell. **×120.**

and the external genitalia. The **ceruminous glands** of the external acoustic meatus and the **apocrine glands of eyelashes (glands of Moll)** are also apocrine-type glands.

Eccrine Sweat Glands

Eccrine sweat glands are simple coiled glands that regulate body temperature.

Eccrine sweat glands are independent structures, not associated with the hair follicle that arises as a downgrowth

from the fetal epidermis. Each eccrine gland is arranged as a blind-ended, simple, coiled tubular structure. It consists of two segments: a **secretory segment** located deep in the dermis or in the upper part of the hypodermis and a directly continuous, less coiled **duct segment** that leads to the epidermal surface (Fig. 15.17 and Plate 44, page 518).

Eccrine sweat glands play a major role in **temperature regulation** through the cooling that results from the evaporation of water from sweat on the body surface. The secretory portion of the glands produces a secretion similar in composition to an ultrafiltrate of blood. Resorption of some of the sodium and water in the duct results in the release of a hypotonic sweat at the skin surface. This hypotonic watery solution is low in protein and contains varying amounts of sodium chloride, urea, uric acid, and ammonia. Thus, the eccrine sweat gland also serves, in part, as an excretory organ.

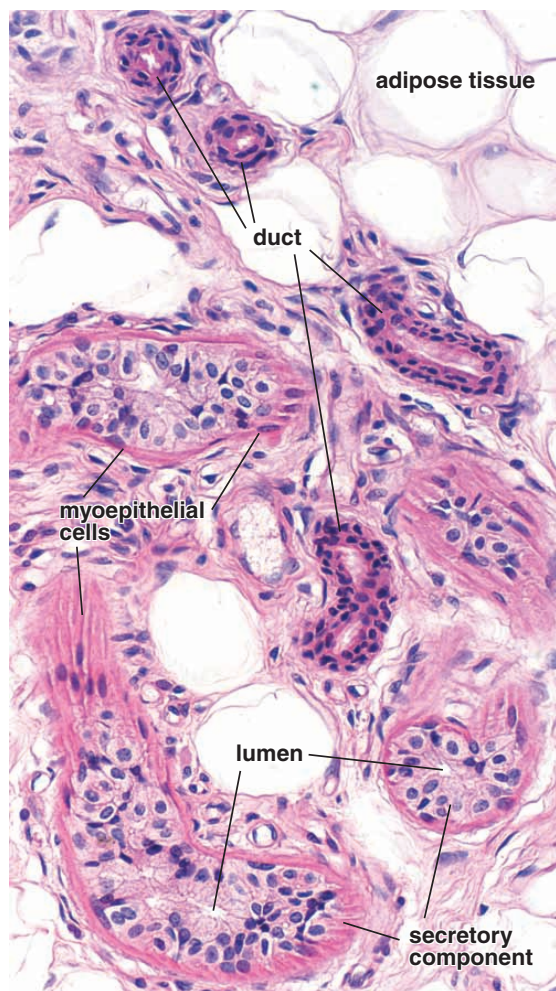


FIGURE 15.17 ▲ Photomicrograph of an eccrine sweat gland.

This photomicrograph of an H&E-stained section of human skin shows profiles of both the secretory component and the duct of an eccrine sweat gland. The secretory component appears as a double layer of cuboidal epithelial cells and peripherally, within the basal lamina, a layer of myoepithelial cells. The duct portion of the gland has a narrower outside diameter and lumen than the secretory portion of the gland. It consists of a double layer of small cuboidal cells without the myoepithelial cells. $\times 320$.

Excessive sweating can lead to the loss of other electrolytes such as potassium and magnesium, as well as significant water loss. Normally, the body loses about 600 ml of water a day through evaporation from the lungs and skin. Under conditions of high ambient temperature, water loss can be increased in a regulated manner by an increased rate of sweating. This **thermoregulatory sweating** first occurs on the forehead and scalp, extends to the face and to the rest of the body, and occurs last on the palms and soles. Under conditions of emotional stress, however, the palms, soles, and axillae are the first surfaces to sweat. Control of thermoregulatory sweating is cholinergic, whereas emotional sweating may be stimulated by adrenergic portions of the sympathetic division of the autonomic nervous system.

The secretory segment of the eccrine sweat gland contains three cell types.

Three cell types are present in the secretory segment of the gland: **clear cells** and **dark cells**, both of which are secretory epithelial cells, and **myoepithelial cells**, which are contractile epithelial cells (Fig. 15.18 and Plate 45, page 520). All of the cells rest on the basal lamina; their arrangement is that of a pseudostratified epithelium.

- **Clear cells** are characterized by abundant glycogen. The glycogen is conspicuous in Figure 15.18a because of its amount; it would stain intensely with the periodic acid–Schiff (PAS) method. In routine H&E preparations, the cytoplasm of clear cells stains poorly. Membranous organelles include numerous mitochondria, profiles of sER, and a relatively small Golgi apparatus. The plasma membrane is remarkably amplified at the lateral and apical surfaces by extensive cytoplasmic folds. In addition, the basal surface of the cell possesses infoldings, although they are considerably less complex than the cytoplasmic folds. The morphology of these cells indicates that they produce the watery component of sweat.
- **Dark cells** are characterized by abundant rER and secretory granules (see Fig. 15.18). The Golgi apparatus is relatively large, a feature consistent with the glycoprotein secretion of these cells. The apical cytoplasm contains mature secretory granules and occupies most of the luminal surface (see Fig. 15.18a). Clear cells have considerably less cytoplasmic exposure to the lumen; their secretion is largely via the lateral surfaces of the cell, which are in contact with intercellular canaliculi that allow the watery secretion to reach the lumen. Here it mixes with the proteinaceous secretion of the dark cells.
- **Myoepithelial cells** are limited to the basal aspect of the secretory segment. They lie between the secretory cells, with their processes oriented transversally to the tubule. The cytoplasm contains numerous contractile filaments (actin) that stain deeply with eosin, thus making them readily identifiable in routine H&E specimens. Contraction of these cells is responsible for rapid expression of sweat from the gland.

The duct segment of eccrine glands is lined by stratified cuboidal epithelium and lacks myoepithelial cells.

The **duct segment** of the gland continues from the secretory portion with coiling. In histologic sections, multiple

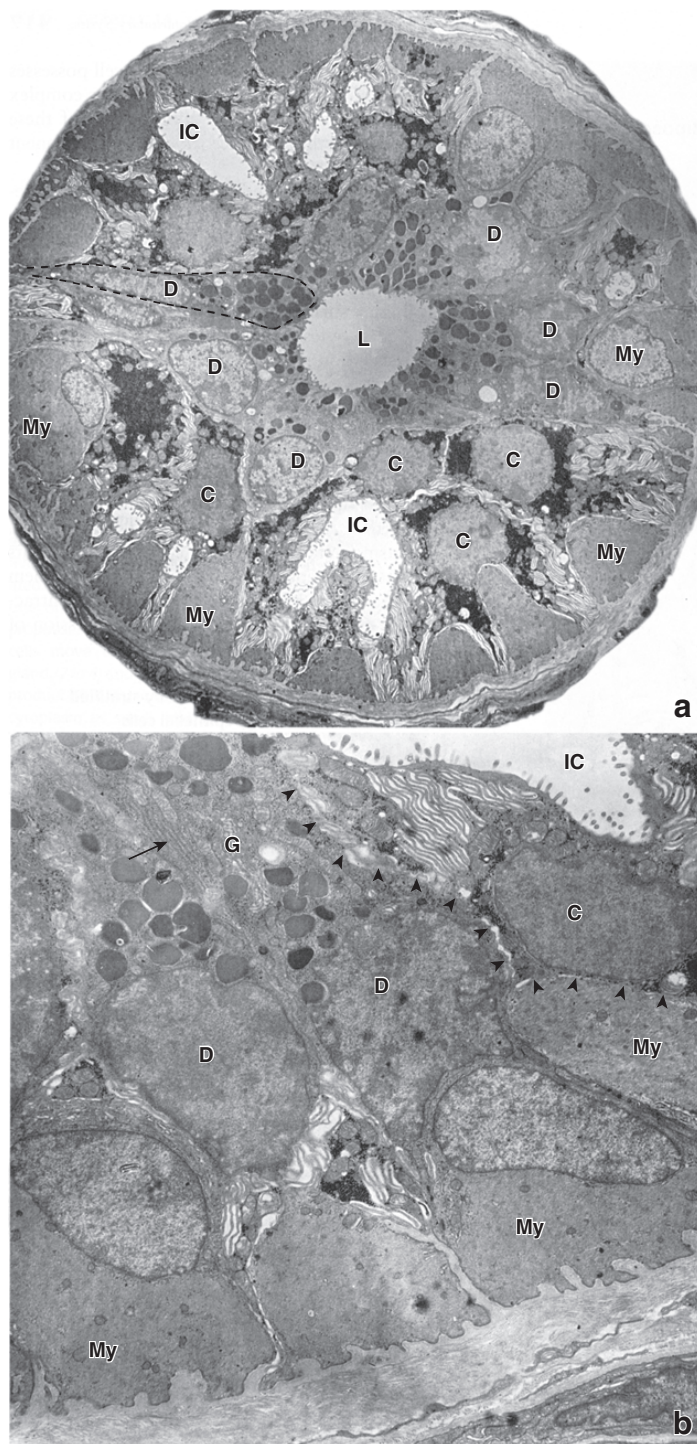


FIGURE 15.18 ▲ Electron micrographs of an eccrine sweat gland. **a.** This micrograph shows myoepithelial cells (*My*) and two distinctive gland cell types, dark cells (*D*) and clear cells (*C*). The apical portion of the dark cell is broad; it faces the lumen (*L*) of the gland and contains numerous secretory granules. The *dashed line* marks the boundary of one dark cell. The clear cell is more removed from the lumen of the gland. Its base rests on the myoepithelial cells or directly on the basal lamina. Most of the free surface of the clear cell faces an intercellular canaliculus (*IC*). Clear cells contain numerous mitochondria, extensive infoldings of the plasma membrane, and large numbers of electron-dense glycogen inclusions. $\times 5,600$. (Courtesy of Dr. John A. Terzakis.) **b.** At higher magnification, dark cells display rER (*arrow*) and a Golgi apparatus (*G*) in addition to secretory granules. Clear cells show large amounts of folded membrane, mitochondria, and glycogen. The myoepithelial cells (*My*) contain large numbers of contractile actin filaments. *Short stubby arrows* (*upper right*) mark the boundary of a clear cell. $\times 17,500$. (Courtesy of Dr. John A. Terzakis.)

duct profiles typically appear among the secretory profiles. As the duct passes upward through the dermis, it takes a gentle spiral course until it reaches the epidermis, where it then continues in a tighter spiral to the surface. When the duct enters the epidermis, however, the duct cells end and the epidermal cells form the wall of the duct. The duct is composed of **stratified cuboidal epithelium**, consisting of a basal cell layer and a luminal cell layer. The duct cells are smaller and appear darker than the cells of the secretory portion of the gland. Also, the duct has a smaller diameter than the secretory portion. In contrast to the secretory portion of the eccrine gland, the duct portion does not possess myoepithelial cells. These features are useful in distinguishing the duct from the secretory portion in a histologic section (see Fig. 15.17).

The basal or peripheral cells of the duct have a rounded or ovoid nucleus and contain a prominent nucleolus. The cytoplasm is filled with mitochondria and ribosomes. The apical or luminal cells are smaller than the basal cells, but their nuclei are similar in appearance. The most conspicuous feature of the luminal cells is the deeply stained, glassy (hyalinized) appearance of the apical cytoplasm. The glassy appearance is attributable to the presence of large numbers of aggregated tonofilaments in the apical cytoplasm.

Apocrine Sweat Glands

Apocrine glands are large-lumen tubular glands associated with hair follicles.

Apocrine sweat glands develop from the same downgrowths of epidermis that give rise to hair follicles. The connection to the follicle is retained, allowing the secretion of the gland to enter the follicle, typically at a level just above the entry of the sebaceous duct. From here, the secretion makes its way to the surface.

Like eccrine glands, apocrine glands are coiled tubular glands. They are sometimes branched. The secretory portion of the gland is located deep in the dermis or, more commonly, in the upper region of the hypodermis.

The secretory portion of apocrine glands has a wider lumen than that of eccrine glands and is composed of a single cell type.

The secretory portion of apocrine glands differs in several respects from that of eccrine glands. The most obvious difference, readily noted in the light microscope, is its very wide lumen (Fig. 15.19 and Plate 44, page 518). Unlike eccrine glands, apocrine glands store their secretory product in the lumen. The secretory portion of the gland is composed of simple epithelium. Only one cell type is present, and the cytoplasm of the cell is eosinophilic. The apical part of the cell often exhibits a bleb-like protrusion. It was once thought that this part of the cell pinched off and was discharged into the lumen as an apocrine secretion, thus the name of the gland. However, TEM studies confirm that the secretion is a **merocrine type**. The apical cytoplasm contains numerous small granules, the secretory components within the cell, which are discharged by exocytosis. Other features of the cell include numerous lysosomes and lipofuscin pigment granules. The latter represent secondary and tertiary lysosomes. Mitochondria are also numerous. During the refractory

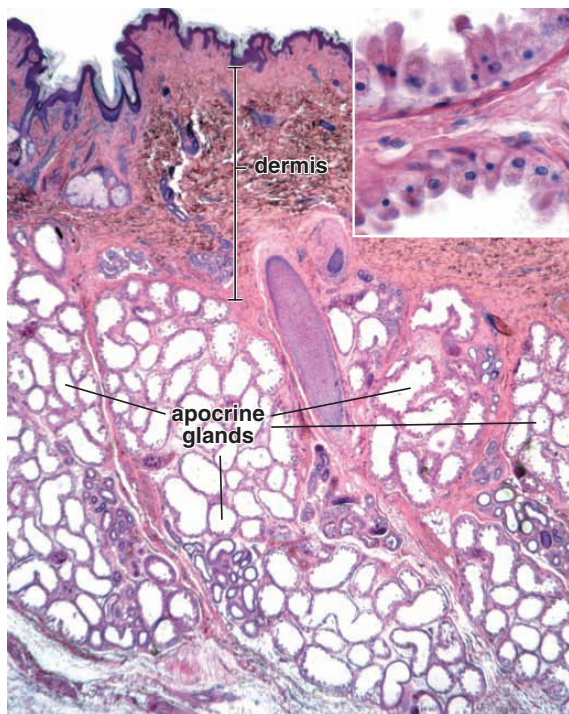


FIGURE 15.19 ▲ Photomicrograph of an apocrine sweat gland. This section of adult skin from the area around the anus shows several apocrine (anal) sweat glands, which are easily identified by the large lumen of their secretory components. This apocrine sweat gland is close to a hair follicle (center of photomicrograph) and deep to the dense, irregular connective tissue of the dermis. **X45. Inset.** Higher magnification of secretory component shows the cell types of the apocrine gland. The gland consists of a simple epithelium whose cells are either cuboidal or columnar and myoepithelial cells located in the basal portion of the epithelial cell layer. **X230.**

phase, after expulsion of the secretion, the Golgi apparatus enlarges in preparation for a new secretory phase.

Myoepithelial cells are also present in the secretory portion of the gland and are situated between the secretory cells and the adjacent basal lamina. As in eccrine glands, contraction of the processes of myoid cells facilitates expulsion of the secretory product from the gland.

The duct portion of apocrine glands is lined by stratified cuboidal epithelium and lacks myoepithelial cells.

The **duct of the apocrine gland** is similar to that of the eccrine duct; it has a narrow lumen. However, it continues from the secretory portion of the gland in a relatively straight path to empty into the follicle canal. Because of its straight course, the probability of viewing both the duct and the secretory portion of an apocrine gland in the same histologic section is reduced. Also in contrast to the eccrine duct, resorption does not take place in the apocrine duct. The secretory product is not altered in its passage through the duct.

The duct epithelium is stratified cuboidal, usually two but sometimes three cell layers thick. The apical cytoplasm of the luminal cells appears hyalinized, a consequence of the aggregated tonofilaments in the apical cytoplasm. In this aspect, they resemble the luminal cells of the eccrine duct.

Apocrine glands produce a protein-rich secretion containing pheromones.

Apocrine glands produce a secretion that contains protein, carbohydrate, ammonia, lipid, and certain organic compounds that may color the secretion. However, the secretions vary with the anatomic location. In the axilla, the secretion is milky and slightly viscous. When secreted, the fluid is odorless; through bacterial action on the skin surface, it develops an acrid odor.

Apocrine glands become functional at puberty; as with axillary and pubic hair, their development depends on sex hormones. In the female, both axillary and areolar apocrine glands undergo morphologic and secretory changes that parallel the menstrual cycle.

In many mammals, similar glands secrete **pheromones**, chemical signals used in marking territory, in courtship behavior, and in certain maternal and social behaviors. It is generally believed that apocrine secretions may function as pheromones in humans. Male pheromones (androstenol and androsteneone) in the secretion of apocrine glands have a direct effect on the female menstruation cycle. Furthermore, female pheromones (copulins) influence male perception of females and may also induce hormonal changes in males.

Both eccrine and apocrine sweat glands are innervated by the sympathetic portion of the autonomic nervous system.

Eccrine sweat glands are stimulated by **cholinergic** transmitters (usually identified with the parasympathetic component of the autonomic system), whereas **apocrine glands** are stimulated by **adrenergic** transmitters. As described above, eccrine glands respond to heat and stress. Apocrine glands respond to emotional and sensory stimuli but not to heat.

Nails

Nails are plates of keratinized cells containing hard keratin.

The slightly arched fingernails and toenails, more properly referred to as **nail plates**, rest on **nail beds**. The nail bed consists of epithelial cells that are continuous with the stratum basale and stratum spinosum of the epidermis (Fig. 15.20 and Plate 47, page 524).

The proximal part of the nail, the **nail root**, is buried in a fold of epidermis and covers the cells of the **germinative zone** or **matrix**. The matrix contains a variety of cells, including stem cells, epithelial cells, melanocytes, Merkel's cells, and Langerhans' cells. The stem cells of the matrix regularly divide, migrate toward the root of the nail, and there differentiate and produce the keratin of the nail. Nail keratin is a **hard keratin**, like that of the hair cortex. Unlike the soft keratin of the epidermis, it does not desquamate. It consists of densely packed keratin filaments embedded in a matrix of amorphous keratin with a high sulfur content, which is responsible for the hardness of the nail. The process of hard keratin formation, as with the hair cortex, does not involve keratohyalin granules. In addition, a cornified cell envelope contains proteins similar to those found in the epidermis.

The constant addition of new cells at the root and their keratinization account for nail growth. As the nail plate grows, it moves over the nail bed. On the microscopic level, the nail

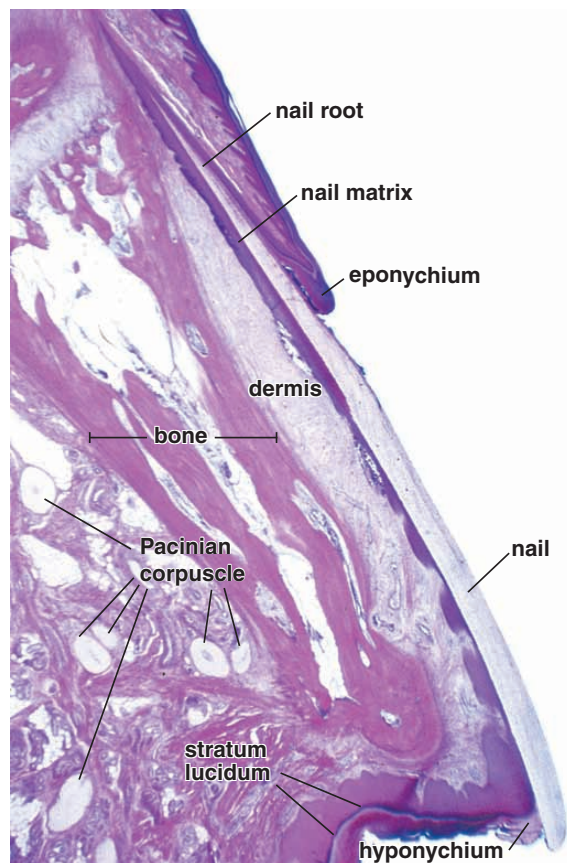


FIGURE 15.20 ▲ Photomicrograph of a sagittal section of distal phalanx with a nail. A nail is a keratinized plate located on the dorsal aspect of the distal phalanges. Under the free edge of the nail is a boundary layer, the hyponychium, which is continuous with the stratum corneum of the adjacent epidermis. The proximal end, the root of the nail, is overlapped by skin, the eponychium, which is also continuous with the stratum corneum of the adjacent epidermis. Deep to the nail is a layer of epithelium with underlying dermis. The proximal portion of this epithelium is referred to as the **nail matrix**. The bone in this section represents a distal phalanx. Numerous Pacinian corpuscles are present in the connective tissue of the palmar side of the finger. Note that even at this low magnification, the stratum lucidum is visible in the epidermis of the fingertip. $\times 10$.

plate contains closely packed interdigitating **corneocytes** lacking nuclei and organelles.

The crescent-shaped white area near the root of the nail, the **lunula**, derives its color from the thick, opaque layer of partially keratinized matrix cells in this region. When the nail plate becomes fully keratinized, it is more transparent and takes on the coloring of the underlying vascular bed. The edge of the skin fold covering the root of the nail is the **eponychium** or **cuticle**. The cuticle is also composed of hard keratin; therefore, it does not desquamate. Because of its thinness, it tends to break off or, as with many individuals, it is trimmed and pushed back. A thickened epidermal layer, the hyponychium, secures the free edge of the nail plate at the fingertip.

The process of wound healing of the skin is classically described as either a primary or secondary union. Healing by **primary union (first intention)** occurs after surgical incisions in which wounds that are usually clean and uninfected have their edges approximated by surgical sutures. The healing by **secondary union (secondary intention)** occurs in traumatic wounds with separated edges, which are characterized by more extensive loss of cells and tissues. Wound healing in such cases involves generating a large amount of **granulation tissue**, which represents a specialized type of tissue formed during the repair process.

The repair of an incision or laceration of the skin requires stimulated growth of both the dermis and the epidermis. Dermal repair involves (1) blood clot formation, (2) removal of damaged collagen fibers, primarily through the effort of macrophage activity that is associated with inflammation, (3) formation of granulation tissue, (4) reepithelialization of the exposed surface, (5) proliferation and migration of fibroblasts and differentiation of myofibroblasts involved in wound contraction, and (6) deposition and remodeling of the extracellular matrix of underlying connective tissue. Healing by primary union following application of sutures reduces the extent of the repair area through maximal closure of a wound, minimizing scar formation. Surgical incisions are typically made along cleavage lines; the cut tends to parallel the collagen fibers, thus minimizing the need for excess collagen production and the inherent scarring that may occur.

Repair of the epidermis involves the proliferation of the basal keratinocytes in the stratum basale in the undamaged site surrounding the wound (Fig. F15.6.1). Mitotic activity is markedly increased within the first 24 hours. In a short time, the wound site is covered by a **scab** that represents dehydrated blood clot. The proliferating basal cells of the stratum basale begin migrating beneath the scab and across the wound surface. The migration rate may be as much as 0.5 mm/day, starting within 8 to 18 hours after wounding. Further proliferation and differentiation occur behind the migration front, leading to restoration of the multilayered epidermis. As new cells ultimately keratinize and desquamate, the overlying scab is freed with the desquamating cells, which explains why a scab detaches from its periphery inward.

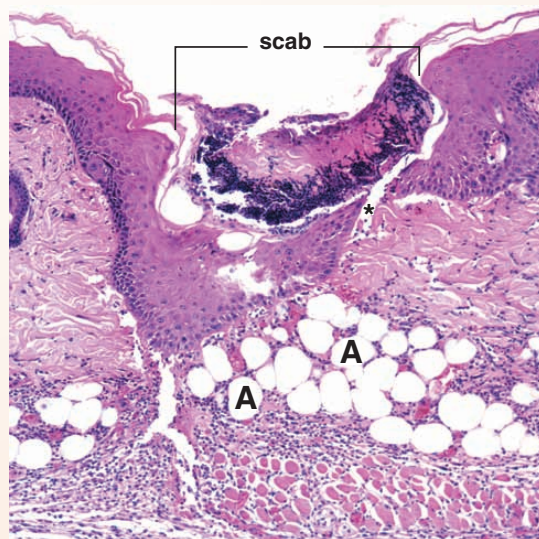


FIGURE F15.6.1 ▲ Photomicrograph showing a late stage in the epidermal repair of a skin wound. The initial injury was caused by an incision through the full thickness of the skin and partially into the hypodermis, which contains adipose cells (A). The epidermis has re-formed beneath the scab. The *asterisk* marks an artifact where epidermis separated during specimen preparation. The scab, which contains numerous dead neutrophils in its inferior aspect, is close to the point of release. The dermis at this stage shows little change during the repair process but will ultimately reestablish itself to form a continuous layer. $\times 110$.

In cases in which the full thickness of the epidermal layer is removed either by trauma or in surgery, the parts of hair follicles, the follicular bulge that contains niche of the epidermal stem cells, will produce cells that migrate over the exposed surface to reestablish a complete epithelial (epidermal) layer. Massive destruction of all of the epithelial structures of the skin, as in a third-degree burn or extensive full-thickness abrasion, prevents reepithelialization. Such wounds can be healed only by grafting epidermis to cover the wounded area. In the absence of a graft, the wound would, at best, reepithelialize slowly and imperfectly by ingrowth of cells from the margins of the wound.



Integumentary System

OVERVIEW OF THE INTEGUMENTARY SYSTEM

- ▶ The **integumentary system** consists of skin and its derivatives (epidermal skin appendages).
- ▶ The **skin** has two layers: the **epidermis**, a superficial layer that consists mainly of a stratified squamous keratinized epithelium; and the **dermis**, a deeper layer of dense irregular connective tissue.
- ▶ Deep to the skin is the **hypodermis**, which contains variable amounts of adipose tissue.

EPIDERMIS

- ▶ The **epidermis** is composed primarily of **keratinocytes** (85%) that undergo differentiation to form stratified squamous keratinized epithelium.
- ▶ Four distinct layers of epidermis (strata) can be distinguished.
- ▶ The **stratum basale** is a single layer of small, mitotically active basal cells that are attached by hemidesmosomes to underlying connective tissue and by desmosomes to each other.
- ▶ The **stratum spinosum** contains several layers of larger keratinocytes that are attached to each other by **desmosomes** located at the ends of their cytoplasmic processes containing intermediate filaments (keratin filaments).
- ▶ The **stratum granulosum** is a distinct layer of flattened keratinocytes filled with **keratohyalin granules** (contain precursors to **filaggrin**, which aggregates **keratin filaments** and lamellar bodies containing lipids, which, when secreted, are responsible for the formation of the epidermal **water barrier**).
- ▶ The **stratum corneum** is the most superficial layer of terminally differentiated squamous cells (with no nuclei) that are entirely filled with keratin filaments. These cells are constantly desquamating from the skin surface.
- ▶ Total epidermal turnover time takes approximately **47 days**.
- ▶ **Melanocytes** (5% of cells in epidermis) reside in the stratum basale and have long processes that extend between keratinocytes into the stratum spinosum.
- ▶ Melanocytes synthesize **melanin** pigment in **melanosomes** and during the process of **pigment donation**, melanocytes transfer them into adjacent keratinocytes. The transferred pigment accumulates above nuclei of keratinocytes to protect nuclear DNA from ultraviolet (UV) radiation and damage.
- ▶ Other cells in the epidermis include **Langerhans' cells** (2% to 5%), which are the antigen-presenting cells involved in signaling of the immune system, and **Merkel's cells** (6% to 10%), which are mechanoreceptor cells associated with sensory nerve endings.

DERMIS

- ▶ The **dermis** is composed of two layers.
- ▶ The **papillary layer** is superficial and consists of loose connective tissue (collagen I and III) that contains extensive plexus of blood, lymphatic vessels, and sensory nerve endings.
- ▶ The **reticular layer** is deeper and is composed of dense irregular connective tissue containing type I collagen, elastic fibers, and larger blood vessels.
- ▶ The **epidermal-dermal junction** has numerous finger-like connective tissue protrusions called **dermal papillae** that correspond to similar epidermal protrusions (**epidermal ridges**).
- ▶ **Dermal papillae** contain nerve endings and a network of blood and lymphatic capillaries.

SENSORY NERVE RECEPTORS OF SKIN

- ▶ The epidermis contains **free nerve endings**, which detect fine touch, heat, cold, and pain.
- ▶ In addition, **Merkel's corpuscle** (Merkel's cell with a nerve ending) is a sensitive mechanoreceptor.
- ▶ The dermis contains several encapsulated nerve endings, such as **Pacinian corpuscles** to detect pressure and vibrations, **Meissner's corpuscles** to detect light touch, and **Ruffini's corpuscles** to detect skin stretch and torque.

EPIDERMAL SKIN APPENDAGES

- ▶ **Hairs** and **hair follicles** are present over almost the entire body.
- ▶ The **hair follicle** contains a reservoir of epidermal stem cells (**follicular bulge**) that are responsible for differentiation into hair-forming matrix cells.
- ▶ **Hair** is formed by the differentiation of **matrix cells** in the **inferior segment** of the hair follicle (**bulb**) to form the medulla, cortex (80% of hair mass), and cuticle of a hair shaft.
- ▶ The **hair shaft** is surrounded by the internal and external root sheath. The **internal root sheath** has three layers of cells: **Henle's layer**, **Huxley's layer**, and the **internal root sheath cuticle**. The **external root sheath** is continuous with the epidermis.
- ▶ **Sebaceous glands** produce **sebum** that coats the hair and skin surface. Sebum is produced by holocrine secretion and is discharged via pilosebaceous canal into the hair follicle.
- ▶ **Apocrine sweat glands** secrete protein-rich sweat into the hair follicles, but they are restricted to specific regions of the body (axillae, perineum).
- ▶ Apocrine sweat glands are coiled tubular glands with wide lumen. Their secretory parts contain **myoepithelial cells**, the contraction of which is responsible for expression of sweat.
- ▶ **Eccrine sweat glands** are not related to hair follicles. They produce sweat that is similar in composition to an ultrafiltrate of blood in kidney.
- ▶ Eccrine sweat glands play a major role in temperature regulation through the cooling that results from the evaporation of water from sweat on the body surface. Their secretory parts also contain myoepithelial cells.
- ▶ **Nails** are plates of keratinized cells resting on **nail beds** containing **hard keratin** that is formed in a nail root at the proximal part of the nail. Keratinocytes proliferate there and differentiate to form hard keratin.
- ▶ As the **nail plate** grows, it moves over the nail bed with edges covered by skin folds.

The **skin**, or integument, consists of two main layers: the **epidermis**, composed of stratified squamous epithelium that is keratinized, and the **dermis**, composed of connective tissue. Under the dermis is a layer of loose connective tissue called the **hypodermis**, which is also generally referred to as the **subcutaneous tissue** or, by gross anatomists, as the superficial fascia. Typically, the hypodermis contains large amounts of adipose tissue, particularly in an adequately nourished individual.

The epidermis gives rise to nails, hairs, sebaceous glands, and sweat glands. On the palms of the hands and soles of the feet, the epidermis has an outer keratinized layer that is substantially thicker than that over the other parts of the body. Accordingly, the skin over the palms and soles is referred to as **thick skin**, in contrast to the skin over other parts of the body, which is referred to as **thin skin**.

There are no hairs in thick skin. In addition, the interface between the epidermis and the dermis is more complex in thick skin than in thin skin. The finger-like projections of the dermis into the base of the epidermis, the **dermal papillae**, are much longer and more closely spaced in thick skin. This provides greater resistance to frictional forces acting on this skin.



Thick skin, human, H&E ×45.

In this sample of thick skin, the **epidermis** (*Ep*) is at the top; the remainder of the field consists of dermis, in which a large number of **sweat glands** (*SW*) can be observed. Although the layers of the epidermis are examined more advantageously at higher magnification, it is easy to see, even at this relatively low magnification, that about half of the thickness of the epidermis consists of a distinctive surface layer that stains more lightly than the remainder of the

epidermis. This is the keratinized layer. The dome-shaped surface contours represent a cross-section through the minute ridges on the surface of thick skin that produce the characteristic fingerprints of an individual.

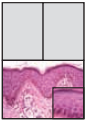
In addition to sweat glands, the dermis displays blood vessels (*BV*) and adipose tissue (*AT*). The ducts of the sweat glands (*D*) extend from the glands to the epidermis. One of the ducts is shown as it enters the epidermis at the bottom of an epithelial ridge. It will pass through the epidermis in a spiral course to open onto the skin surface.



Thin skin, human, H&E ×60.

A sample of thin skin is shown here to compare with the thick skin in the above figure. In addition to sweat glands, thin skin contains hair follicles (*HF*) and their associated **sebaceous**

glands (*SGI*). Each sebaceous gland opens into a hair follicle. Often, as in this tissue sample, the hair follicles and the glands, both sebaceous and sweat, extend beyond the dermis (*De*) and into the hypodermis. Note the blood vessels (*BV*) and adipose tissue (*AT*) in the hypodermis.



Epidermis, skin, human, H&E ×320; inset ×640.

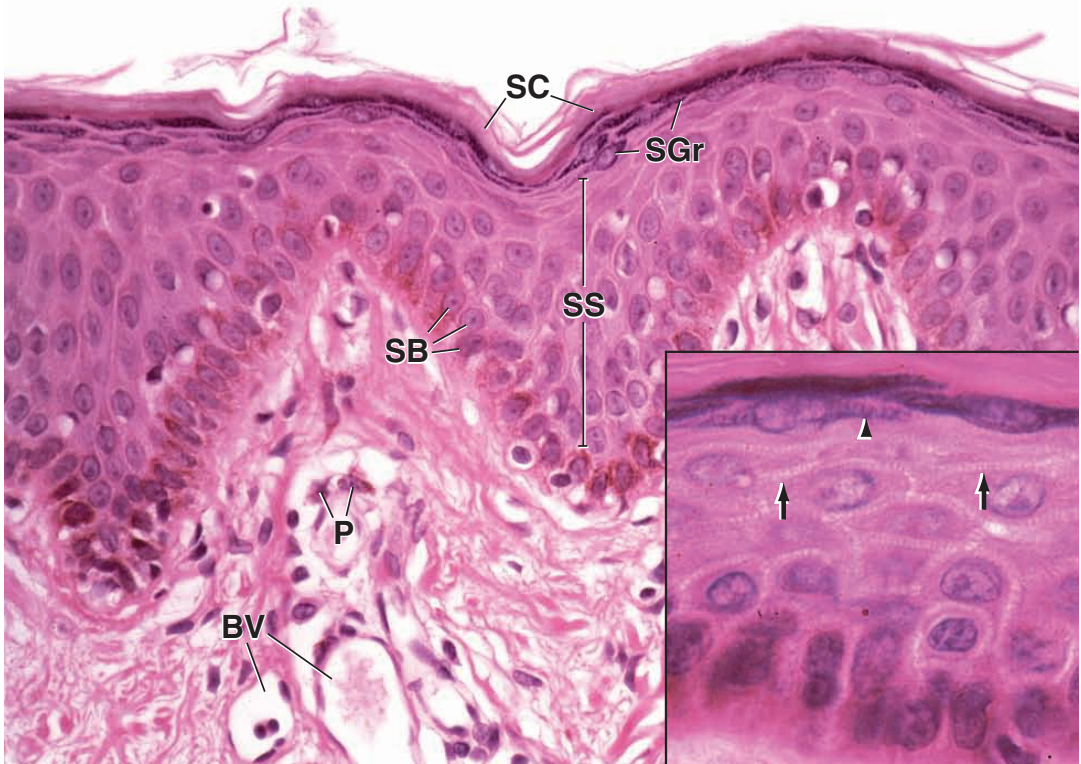
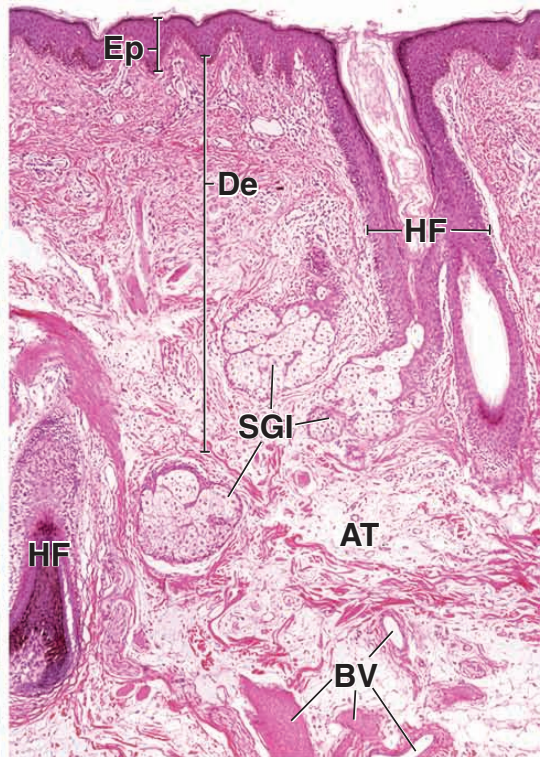
The layers of the **epidermis of thin skin** are shown here at higher magnification. The cell layer that occupies the deepest location is the **stratum basale** (*SB*). This is one cell deep. Just above this is a layer several cells in thickness, the **stratum spinosum** (*SS*). It consists of cells that have spinous processes on their surface. These processes meet with spinous processes of neighboring cells and, together, appear as intercellular bridges (*arrows*, *inset*). The next

layer is the **stratum granulosum** (*SGr*), whose cells contain keratohyalin granules (*arrowhead*, *inset*). On the surface is the **stratum corneum** (*SC*). This consists of keratinized cells, that is, cells that no longer possess nuclei. The keratinized cells are flat and generally adhere to other cells above and below without evidence of cell boundaries. In thick skin, a fifth layer, the **stratum lucidum**, is seen between the stratum granulosum and the stratum corneum. The pigment in the cells of the stratum basale is melanin; some of this pigment (*P*) is also present in connective tissue cells of the dermis.

AT, adipose tissue
BV, blood vessels
D, duct of sweat glands
De, dermis
Ep, epidermis
HF, hair follicle

P, pigment
SB, stratum basale
SC, stratum corneum
SGI, sebaceous gland
SGr, stratum granulosum
SS, stratum spinosum

SW, sweat gland
arrowhead, granules in cell of stratum granulosum
arrows, intercellular bridges



The **epidermis** contains four distinctive cell types: **keratinocytes**, **melanocytes**, **Langerhans' cells**, and **Merkel's cells**. Keratinocytes are the most numerous of these cells; they are generated in the **stratum basale** and move toward the surface. As they do so, they produce the intracellular protein **keratin** and the special extracellular lipid that serves as a water barrier in the upper layers of the epidermis. Histologically, the keratinocytes are the cells that show spinous processes in the **stratum spinosum**. The other three cell types are not readily identified in H&E-stained paraffin sections. The product of the melanocyte is, however, evident in H&E sections, and this is considered in the first two figures of this plate.

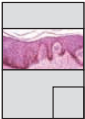
The skin contains a pigment, **melanin**, which protects the tissue against the harmful effects of ultraviolet light. It is formed by the melanocytes that then pass the pigment to the keratinocytes. More pigment is present in dark skin than in light skin; this can be seen by comparing light skin (*top figure*) and dark skin (*middle figure*). The epidermis and a small amount of the dermis are shown in each figure. Whereas the deep part of the dark skin contains considerable pigment, the amount of pigment in light skin is insufficient to be noticeable at this magnification. Cells for producing the pigment are present in both skin types and in approximately equal numbers. The difference is due to more rapid digestion of the pigment by lysosomes of keratinocytes in light skin. After prolonged exposure to sunlight, pigment is also produced in sufficient amounts to be seen in light skin.



Light skin, human, H&E $\times 300$.

In routine H&E-stained paraffin sections of light skin, such as this sample, the **melanocytes** are among the cells that appear as small, rounded, clear cells (CC) mixed with the other cells of the stratum basale. However, not

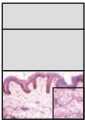
all clear cells of the epidermis are melanocytes. For example, Langerhans' cells may also appear as clear cells, but they are located more superficially in the stratum spinosum. Merkel's cells may also appear as clear cells, thus making it difficult to identify these three cell types with certainty.



Dark skin, human, H&E $\times 300$.

In dark skin, most of the pigment is in the basal portion of the epidermis, but it is also present in cells progressing toward the surface and within the nonnucleated cells of the keratinized

layer. The *arrows* indicate the **melanin pigment** in keratinocytes of the stratum spinosum and in the stratum corneum. In light skin, the melanin is broken down before it leaves the upper part of the stratum spinosum. Thus, pigment is not seen in the upper layers of the epidermis.



Dermis, skin, human, H&E and elastin stain $\times 200$; inset $\times 450$.

This figure is included because it shows certain features of the dermis, the connective tissue layer of the skin. The dermis is divided into two layers: the **papillary layer (PL)** of loose connective tissue and the **reticular layer (RL)** of more dense connective tissue. The papillary layer is immediately under the epidermis. It includes the connective tissue papillae that project into the undersurface of the epidermis. The **reticular layer** is deep to the papillary layer. The boundary between these two layers is not demarcated by any specific structural feature except for the change in the histologic composition of the two layers.

This specimen was stained with H&E and also with a procedure to display elastic fibers (EF). They are relatively thick and conspicuous in the reticular layer (see also *inset*), where they appear as the dark-blue profiles, some of which are elongate, whereas others are short. In the papillary layer, the elastic fibers are thinner and relatively sparse (*arrows*). The *inset* shows the typical eosinophilic staining of the thick collagenous fibers in the reticular layer. Although the collagenous fibers at the lower magnification of this figure are not as prominent, it is nevertheless possible to note that they are thicker in the reticular layer than in the papillary layer. The papillary layer is evidently more cellular than the reticular layer. Many of the small dark-blue profiles in the reticular layer represent oblique and cross-sections through elastic fibers (see *inset*) and not nuclei of cells.

CC, clear cells
EF, elastic fibers

PL, papillary layer
RL, reticular layer

arrows: middle figure, pigment in different layers of epidermis; lower figure, delicate elastic fibers

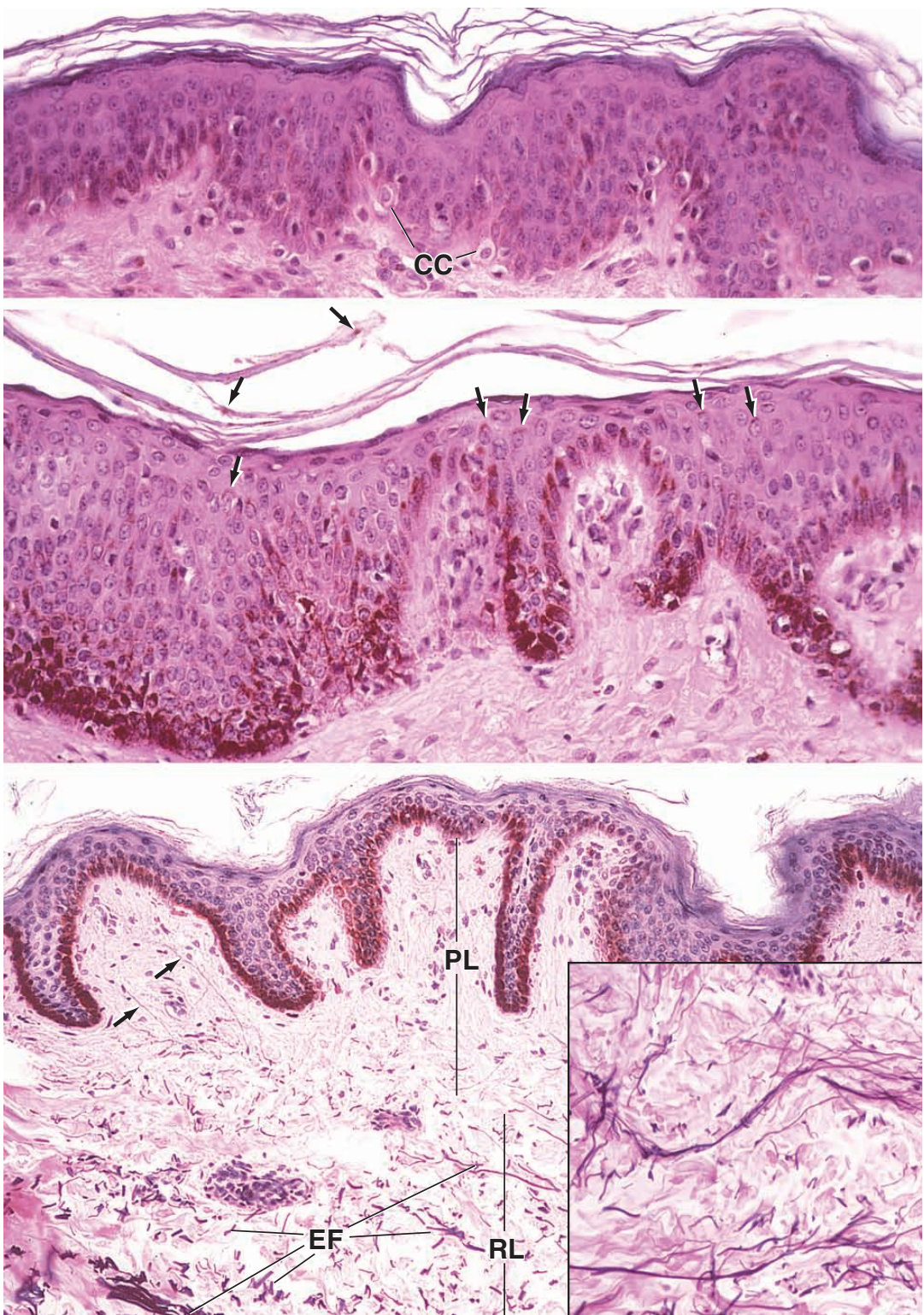
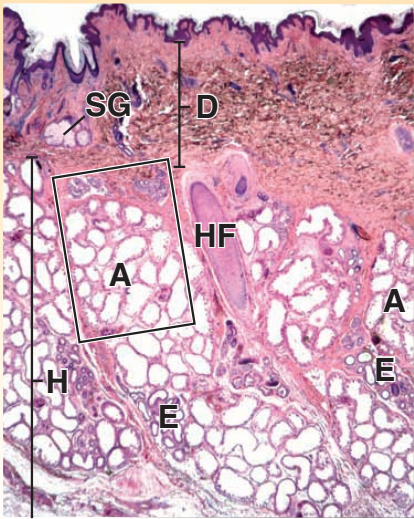


PLATE 44 Apocrine and Eccrine Sweat Glands

Skin possesses three types of glands: **eccrine**, **apocrine**, and **sebaceous**. Eccrine sweat glands are distributed over the entire body surface except for the lips, glans penis, prepuce, clitoris, and labia minora. They are especially numerous in the thick skin of the hands and feet. Evaporation of the secreted sweat on the skin surface cools the body.

Apocrine sweat glands are localized in the axilla, areolae, perineal and perianal area, prepuce, scrotum, mons pubis, and labia majora. Many of the epithelial cells in the secretory segment of these glands exhibit an apical bleb-like protrusion that was earlier thought to represent their mode of secretion (i.e., pinching off of the bleb as the secretory product, thus the name apocrine). It is now known that secretion occurs as a merocrine process. The secretion is a clear, viscous product that becomes odiferous through the action of resident microbes on the skin surface. In the human, its role is unclear, but it is generally believed that the secretion may act as a sex attractant (pheromone). Apocrine glands are present at birth but do not fully develop and become functional until puberty. In the female, these glands undergo changes that parallel the menstrual cycle.

ORIENTATION MICROGRAPH: The adjacent orientation micrograph of the skin of the axilla shows both the large, branched tubular apocrine glands (**A**) and the smaller, simple tubular eccrine glands (**E**) in the hypodermis (**H**). Also evident is a tangentially cut hair follicle (**HF**). The overlying dermis (**D**) consists of dense connective tissue and includes part of a sebaceous gland (**SG**).



Apocrine sweat gland, skin, human, H&E $\times 33$.

Low-power micrograph showing the secretory segment of an **apocrine sweat gland**. The sectioned profiles seen here represent several coiled branches of a single gland

surrounded by dense connective tissue (**DCT**). In the *upper part* of this image are two **sweat glands** (**SwG**) also surrounded by dense connective tissue. Note the considerable difference in diameter and lumen size of the two types of glands.



Apocrine sweat gland, skin, human, H&E $\times 256$.

The epithelium (**Ep**) of the **apocrine sweat gland** from the boxed area to the left is simple columnar. The individual cells vary in height, and some show the bleb-like

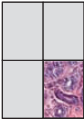
protrusions (**B**). At the base of the epithelium are the spindle-shaped myoepithelial cells. In some regions of the tubule, these cells have been cut longitudinally and thus appear as a deeply stained eosin band (**EB**). At other sites, the cells have been sectioned tangentially and appear as a series of parallel linear profiles (**MyC**).



Eccrine sweat gland, skin, human, H&E $\times 256$.

In this micrograph, the **eccrine sweat gland** from above is seen at higher magnification. Both secretory and duct segments are present. The **secretory segment** (**SS**)

has a wider diameter and larger lumen than the duct segment (**DS**). The epithelium of the secretory segment is simple columnar; the **duct segment** is two cell layers thick, namely, stratified cuboidal. Also, the secretory segment possesses a myoepithelial component.



Eccrine sweat gland, skin, human, H&E $\times 512$.

At this very high magnification, two cross-sectioned profiles of the secretory segment (**SS**) and one profile of the duct segment (**DS**) from the boxed area to the left are shown. When the tubule wall of the secretory segment is cut in a perpendicular plane, the simple columnar nature of the epithelium (**Ep**) is evident. Because

the tubule is so tortuous, more often the epithelium appears to be multi-layered. The **myoepithelial cells** of the secretory segment appear here both as a circumferential band (**CB**) and in a cross-sectioned array (**CA**) in which they resemble the teeth of a saw blade. Occasionally, myoepithelial cell nuclei (**MyN**) are present. Such profiles give the appearance of a pseudostratified epithelium. The duct segment (**DS**) lacks the myoepithelium and also differs in that it is stratified cuboidal. See next plate.

A, apocrine gland
B, bleb-like protrusions
CA, cross-sectioned array
CB, circumferential band
D, dermis
DCT, dense connective tissue

DS, duct segment
E, eccrine gland
EB, eosin band
Ep, epithelium
H, hypodermis
HF, hair follicle

MyC, linear profiles of myoepithelial cells
MyN, myoepithelial cell nuclei
SG, sebaceous gland
SS, secretory segment
SwG, sweat glands

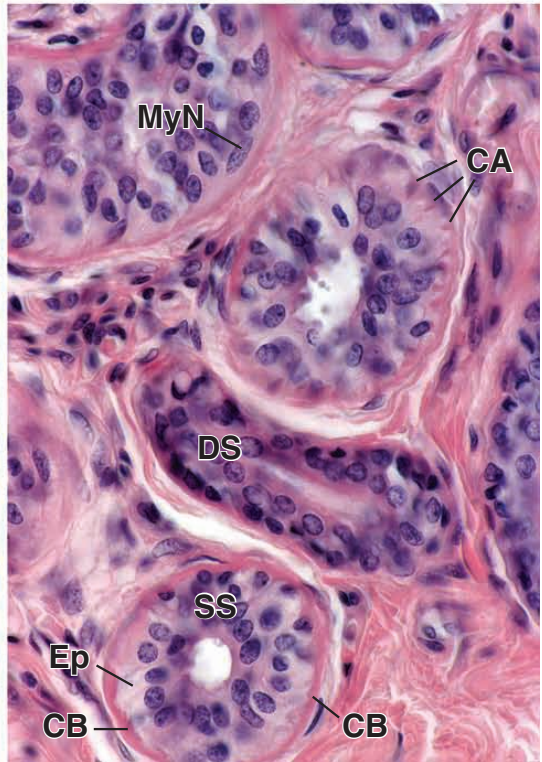
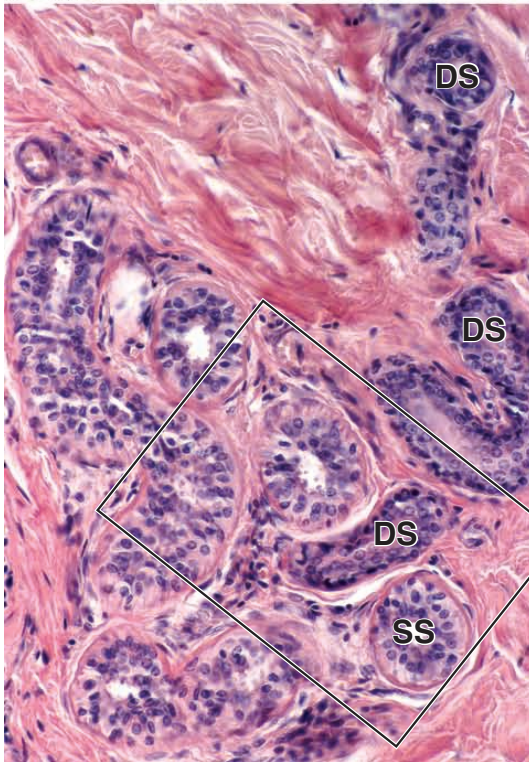
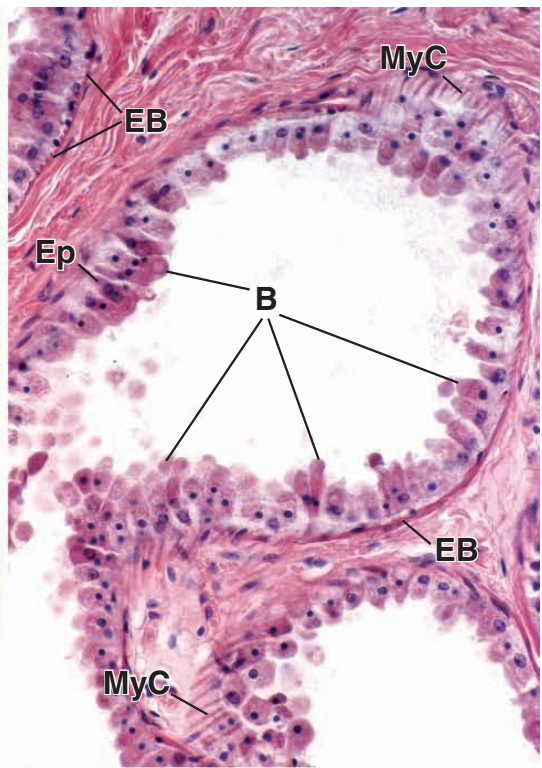
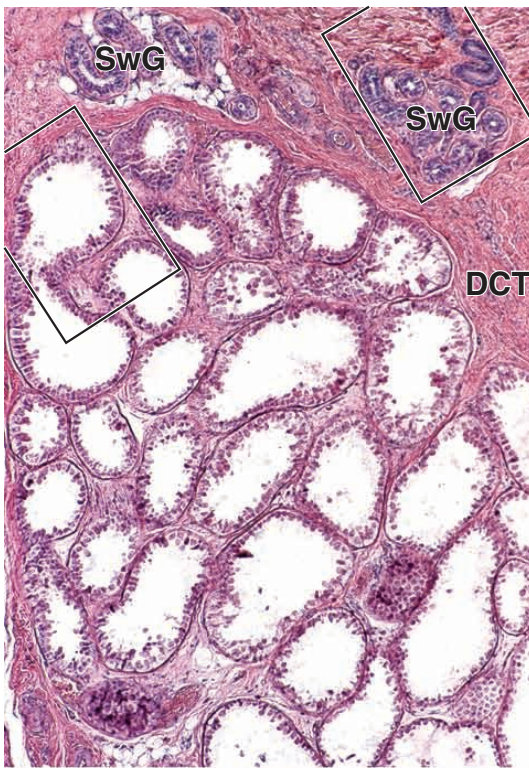
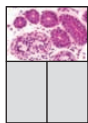


PLATE 45 Sweat and Sebaceous Glands

Normally, the body loses ~600 mL of water a day through evaporation from the lungs and skin. Under conditions of high ambient temperature, water loss is increased by an increased rate of sweating. This *thermoregulatory sweating* first occurs on the forehead and scalp, extends to the face and the rest of the body, and occurs last on the palms and soles. *Emotional sweating*, however, occurs first on the palms and soles and in the axillae. Sweating is under both nervous control through the autonomic nervous system and hormonal control.

Sebaceous glands secrete sebum, an oily substance that coats the hair and skin surface. Sebaceous secretion is a *holocrine* secretion; the entire cell produces, and becomes filled with, the fatty secretory product while it simultaneously undergoes progressive disruption, followed by apoptosis, as the product fills the cell. Both secretory product and cell debris are discharged into the **pilosebaceous canal**.



Sweat gland, skin, human, H&E $\times 1,000$.

This section through a **sweat gland** shows five profiles of the ductal portion (*D*) and two profiles of the secretory portion (*SG*). The larger secretory segment is through a region either just below or above where a U turn was made; therefore, it shows two luminal profiles. The lumina of both the ductal and the secretory units are marked by *asterisks*.

The **glandular unit** of the eccrine sweat gland contains two epithelial cell types and myoepithelial cells (*M*). *Arrowheads* show small cross-sections of myoepithelial cell cytoplasm; *large arrows* show where

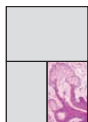
more elongate profiles of myoepithelial cytoplasm are evident. The epithelial cells are of two types, designated dark cells and clear cells. Unfortunately, the characteristic dark cytoplasmic staining of the dark cells is not evident unless special precautions are taken to preserve the secretory granules in their apical cytoplasm. Nevertheless, note that the dark cells are closer to the lumen, whereas the clear cells are closer to the base of the epithelial layer, making contact with either the basal lamina or, more frequently, the myoepithelial cells. In addition, the clear cells are in contact with intercellular canaliculi. Several such intercellular canaliculi are shown in the secretory units (*small arrows*). This figure also shows that the duct consists of two layers of small cuboidal cells.



Sebaceous gland, skin, human, H&E $\times 160$.

Sebaceous glands develop from the epithelial cells of the hair follicle and discharge their secretion into the follicle, from where it reaches the skin surface. The sebaceous secretion is rich in lipid, and this is reflected in the cells of the sebaceous gland. A section of a sebaceous gland and its related hair follicle is shown in

this figure. At this level, the hair follicle consists of the external root sheath (*RS*) surrounding the hair shaft. The **sebaceous gland** (*Seb*) appears as a cluster of cells, most of which display a washed-out or finely reticulated cytoplasm. This is because these cells contain numerous lipid droplets and the lipid is lost by dissolution in fat solvents during the routine preparation of the H&E-stained paraffin section. The opening of the sebaceous gland through the external root sheath (*eRS*) and into the hair follicle is shown in the *lower right*.



Sebaceous gland, skin, human, H&E $\times 280$.

The **sebaceous gland** and **pilosebaceous canal** are shown here at higher magnification. *Numbers 1 to 4* show a series of sebum-producing cells filled with an increasingly

greater amount of lipid and progressively closer to the opening of the pilosebaceous canal (*PSC*) as it enters into the hair follicle. The sebaceous secretion includes the entire cell, and therefore, cells need to be replaced constantly in the functional gland. Cells at the periphery of the gland are basal cells (*BC*). Dividing cells in the basal layer replace those that are lost with the secretion.

BC, basal cells
CT, connective tissue
D, duct of eccrine sweat gland
eRS, junction between sebaceous gland and external root sheath
M, myoepithelial cell
PSC, pilosebaceous canal

RS, external root sheath of hair follicle
Seb, sebaceous gland
SG, secretory component of eccrine sweat gland
arrowheads, myoepithelial cell cytoplasm (cross-section)

asterisks, lumina of glands and ducts
large arrows, myoepithelial cell cytoplasm (longitudinal section)
numbers 1 to 4 (lower right image), see text
small arrows, intercellular canaliculi

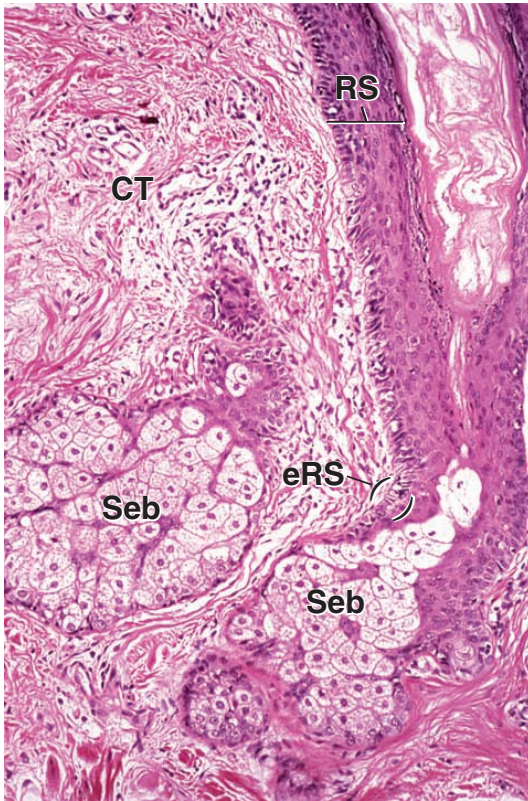
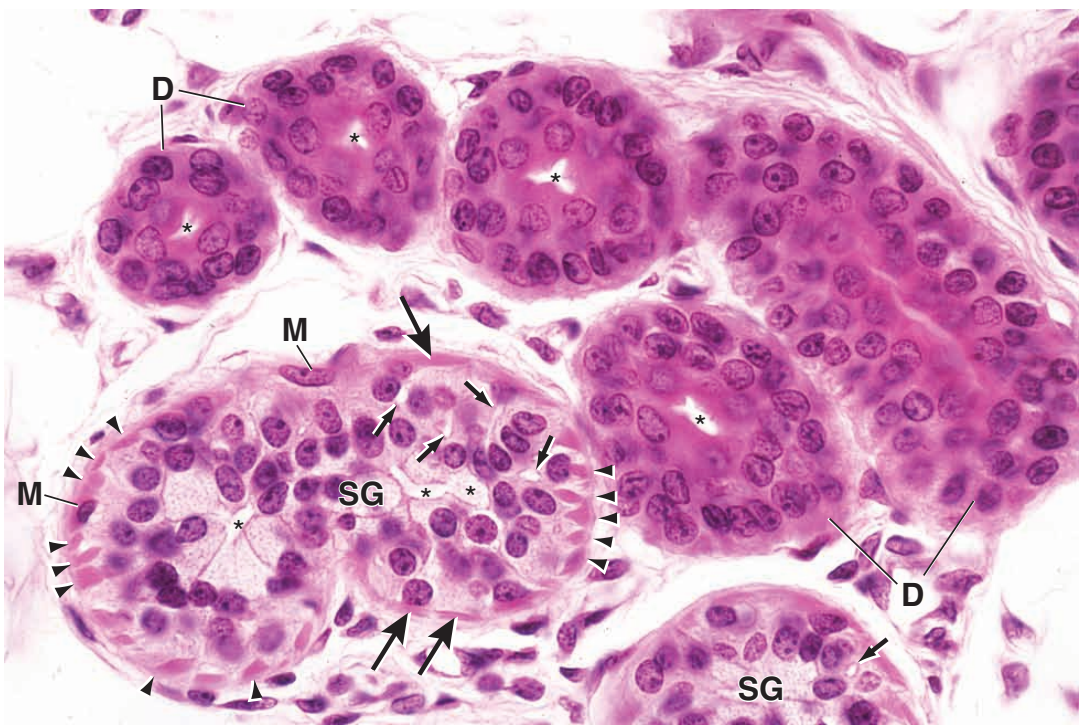


PLATE 46 Integument and Sensory Organs

The **skin** is endowed with numerous **sensory receptors** of various types. These are the peripheral terminals of sensory nerves whose cell bodies are in the dorsal root ganglia. The receptors in the skin are described as **free nerve endings** and **encapsulated nerve endings**. Free nerve endings are the most numerous. They subserve fine touch, heat, and cold and are found in the basal layers of the epidermis and as a network around the root sheath of hair follicles. Encapsulated nerve endings include **Pacinian corpuscles** (deep pressure), **Meissner's corpuscles** (touch, especially in the lips and thick skin of fingers and toes), and Ruffini endings (sustained mechanical stress on the dermis).

Motor endings of the autonomic nervous system supply the blood vessels, the arrector pili muscles, and the apocrine and eccrine sweat glands.



Skin, fingertip, human, H&E $\times 20$.

This specimen is a section of thick skin from the fingertip, showing the **epidermis** (*Ep*) and the **dermis** (*De*) and, under the skin, a portion of the **hypodermis** (*Hy*). The thickness of the epidermis is largely due to the thickness of the stratum corneum. This layer is more lightly stained than the deeper portions of the epidermis. Note, even at this low magnification, the thick collagenous fibers in the reticular layer of the dermis. **Sweat glands** (*SG*) are present in the upper part of the hypodermis, and several sweat ducts (*D*) are seen passing through the epidermis. A feature of this specimen is that it depicts those sensory receptors that can be

recognized in a routine low-power H&E-stained paraffin section. They are Meissner's corpuscles and Pacinian corpuscles (*PC*). Several nerve bundles (*N*) are seen in proximity to the Pacinian corpuscles. Meissner's corpuscles are in the upper part of the dermis, in the dermal papillae immediately under the epidermis. These corpuscles are small and difficult to identify at this low magnification; however, their location is characteristic. Knowing where they are located is a major step in finding Meissner's corpuscles in a tissue section; they are shown at higher magnification in the figure below.

Pacinian corpuscles are seen in the lower part of the hypodermis. These corpuscles are large, slightly oval structures, and even at low magnification, a layered or lamellated pattern can be discerned.



Pacinian corpuscle, skin, human, H&E $\times 320$.

At this higher magnification, the concentric layers or lamellae of the **Pacinian corpuscle** can be seen to be due to flat cells. These are fibroblast-like cells, and although not evident within the tissue section, these cells are continuous with the

perineurium of the nerve fiber. The space between the cellular lamellae contains mostly fluid. The neural portion of the Pacinian corpuscle travels longitudinally through the center of the corpuscle. In this specimen, the corpuscle has been cross-sectioned; an **arrowhead** points to the centrally located nerve fiber.



Meissner's corpuscle, skin, human, H&E $\times 190$.

This high-magnification micrograph shows portions of the *upper left* field of the figure above in which two **Meissner's corpuscles** (*MC*) are in direct proximity to the undersurface of the epidermis in adjacent dermal papillae.

The section shows the long axis of the corpuscles. A Meissner's corpuscle consists of an axon (sometimes two) taking a zigzag or flat spiral course from one pole of the corpuscle to the other. The nerve fiber terminates at the superficial pole of the corpuscle. Consequently, as seen here, the nerve fibers and supporting cells are oriented approximately at right angles to the long axis of the corpuscle. Meissner's corpuscles are particularly numerous near the tips of the fingers and toes.



Meissner's corpuscle, skin, human, H&E $\times 550$.

At the even higher magnification of this figure, the close apposition of **Meissner's corpuscle** to the undersurface

of the epidermis is well demonstrated throughout the entire area of the dermal papilla. The flat spiral path of the neuron (not seen) and its supporting cells is evident here, as is the fibrous capsule (*FC*) that surrounds the ending.

D, ducts of sweat glands
De, dermis
Ep, epidermis
FC, fibrous capsule

Hy, hypodermis
MC, Meissner's corpuscles
N, nerve bundles
PC, Pacinian corpuscles

SG, sweat glands
arrowhead, nerve fiber in the center of Pacinian corpuscle

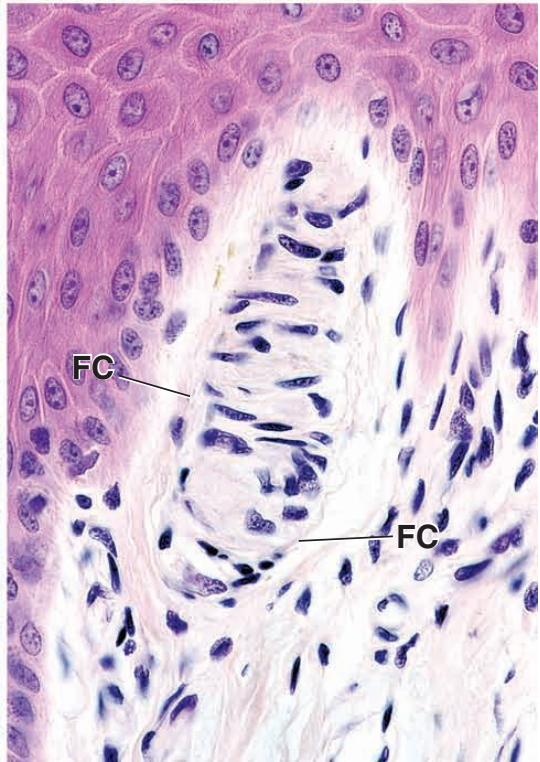
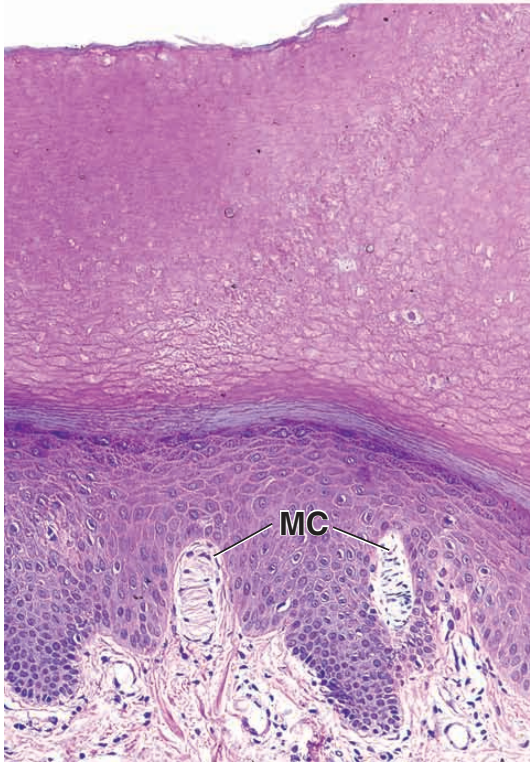
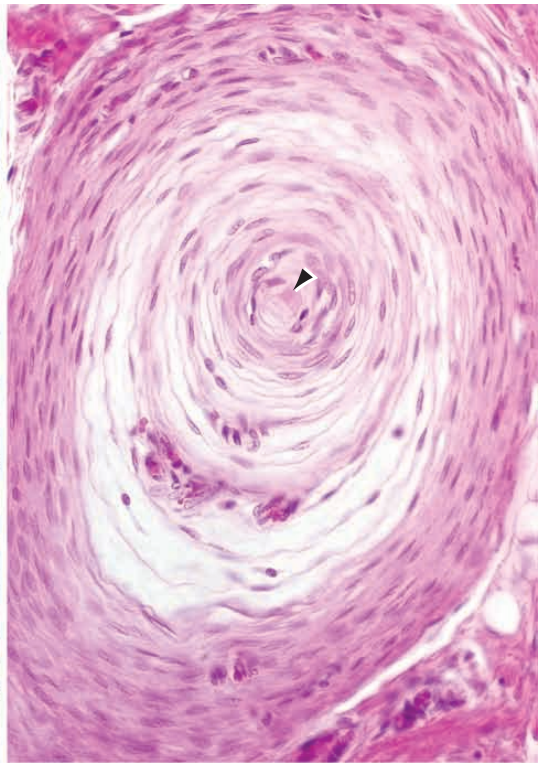
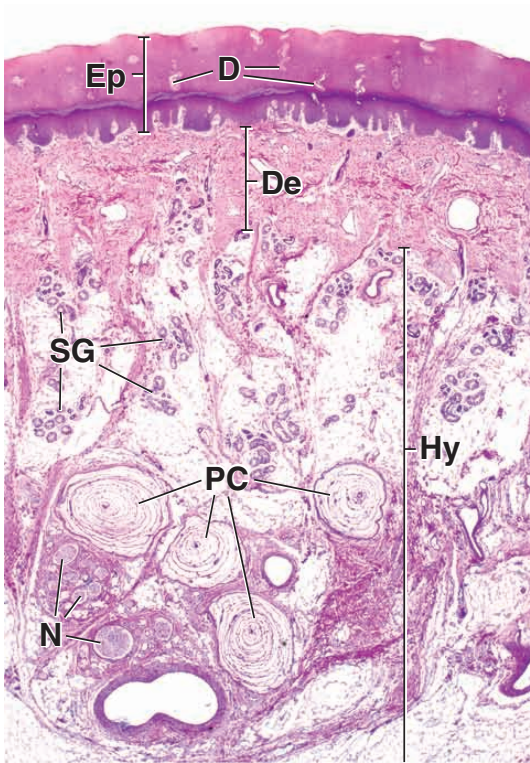
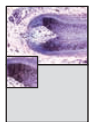


PLATE 47 Hair Follicle and Nail

Hairs are composed of keratinized cells that develop from hair follicles. Hairs are present over almost the entire body, being conspicuously absent only from the sides and palmar surfaces of the hands, from the sides and plantar surfaces of the feet, from the lips, and from the skin around the urogenital orifices. Coloration of the hair is due to the content and type of melanin that it contains. The follicle varies in appearance, depending on whether it is in a growing or a resting phase; the growing follicle is the more elaborate.

The **skin appendages** (adnexa), especially hair follicles and sweat glands, are particularly important in healing of skin wounds. They serve as the source of new epithelial cells when there is extensive loss of epidermis, as in deep abrasions and second-degree burns.



Hair follicle, skin, human, H&E $\times 300$; inset $\times 440$.

The growing end of a **hair follicle** consists of an expanded bulb of epithelial cells that is invaginated by a papilla (*HP*) of connective tissue. The epithelial cells surrounding the papilla at the very tip of the follicle are not yet specialized; they constitute the matrix, the region of the hair follicle where cell division occurs. As the cells leave the matrix, they form cell layers that will become the shaft of the hair and the internal and external root sheaths of the hair follicle.

The cells that will develop into the **shaft of the hair** are seen just to the right of the expanded bulb. They constitute the cortex (*C*), medulla (*M*), and cuticle (*asterisks*) of the hair. The cells of the cortex become keratinized. This layer will come to constitute most of the hair shaft as a thick cylinder. The medulla forms the centrally located axis of the hair shaft; it does not always extend through the entire length of the hair and is absent from some hairs. The cuticle consists of overlapping

cells that ultimately lose their nuclei and become filled with keratin. The cuticle covers the hair shaft like a layer of overlapping shingles.

The **root sheath** (*RS*) has two parts: the external root sheath, which is continuous with the epidermis of the skin, and the internal root sheath, which extends only as far as the level at which sebaceous glands enter the hair follicle. The internal root sheath is further divided into three layers: Henle's layer, Huxley's layer, and the cuticle of the internal root sheath. These layers are seen in the growing hair follicle and are shown at higher magnification in the *inset* with *numbers 1 to 5*: 1, cells of the external root sheath; 2, Henle's layer; 3, Huxley's layer; 4, cuticle of the internal root sheath; and 5, future cuticle of the hair.

Many of the cells of the growing hair follicle contain pigment that contributes to the color of the hair. Most of this pigment is inside the cell (*inset*); however, in very dark hair, some pigment is also extracellular.

The connective tissue surrounding the hair follicle forms a distinct layer referred to as the *sheath*, or **dermal sheath** (*DS*), of the hair follicle.



Nail, skin, human, H&E $\times 12$.

A **nail** is a keratinized plate located on the dorsal aspect of the distal phalanges. A section through a nail plate is shown here. The **nail** itself (*N*) is difficult to stain. Under the free edge of the nail is a boundary layer, the **hyponychium** (*Hypon*), which is continuous with the stratum corneum of the adjacent epidermis. The proximal end of the nail is overlapped by skin; here, the junctional region is called the **eponychium** (*Epon*) and is also continuous with the stratum corneum of the adjacent epidermis. Under the nail is a layer of epithelium, the proximal portion of which is referred to as the **nail matrix** (*NM*). The cells of the nail matrix function in

the growth of the nail. Together, the epithelium under the nail and the underlying dermis (*D*) constitute the nail bed. The proximal portion of the nail, covered by the fold of the skin, is the root of the nail (*NR*).

The relationship of the nail to other structures in the fingertip is also shown in this figure. The bone (*B*) in the specimen represents a distal phalanx. Note that in this bone, there is an epiphyseal growth plate (*EP*) at the proximal extremity of the bone but not at the distal extremity. Numerous Pacinian corpuscles (*PC*) are present in the connective tissue of the palmar side of the finger. Also seen to advantage in this section is the stratum lucidum (*SL*) in the epidermis of the thick skin of the fingertip.

B, bone
C, cortex
D, dermis
DS, dermal sheath
EP, epiphyseal plate
Epon, eponychium
HP, dermal papilla of hair follicle

Hypon, hyponychium
M, medulla
N, nail or nail plate
NM, nail matrix
NR, nail root
PC, Pacinian corpuscles
RS, root sheath

SL, stratum lucidum
asterisks, cuticle of hair
numbers: 1, external root sheath;
2, Henle's layer; 3, Huxley's layer;
4, cuticle of internal root sheath;
5, future cuticle of the hair

